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Integration of machine learning with comprehensive IVIF-QFD-MCDM framework for enhancing online hotel operations

Peide Liu [a,b,*], Xinming Shi [a], Yingcheng Xu [a], Ran Dang [a], Yifan Wu [a]

[a] School of Management Science and Engineering, Shandong University of Finance and Economics, Jinan, China
[b] Shandong Key Laboratory of Blockchain Finance, Shandong University of Finance and Economics, Jinan 250014, China

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ABSTRACT

As online hotel booking becomes increasingly mainstream, the significance of online reviews has escalated, making them pivotal references for potential customers. Consequently, hoteliers must prioritize understanding and leveraging these reviews to enhance their operational performance, which directly correlates with market share and customer satisfaction. Improving operational performance represents a complex multi-criteria decision-making challenge that necessitates a systematic approach. First of all, this paper proposes an innovative approach that integrates Quality Function Deployment (QFD) with multi-criteria decision making to effectively translate customer requirements (CRs) into actionable product characteristics (PCs). Secondly, to address the inherent uncertainties and preference disparities in hotel reviews, we employ an interval-valued intuitionistic fuzzy (IVIF)-VIekriterijumsko KOmpromisno Rangiranje (VIKOR) method for prioritizing the hotel operation measures. Furthermore, we harness the capabilities of deep learning to extract and analyze CRs from online reviews. Specifically, we utilize the Latent Dirichlet Allocation (LDA) model to identify relevant topics within the reviews and the Robustly Optimized BERT Pre-Training Approach (RoBERTa) to assess the sentiment associated with these topics. More importantly, the weighting of CRs and PCs is determined using a combined approach of importance and performance analysis (IPA) and a combined Kano model (CKM), facilitated by an IVIF-QFD matrix. This comprehensive model enables hoteliers to quickly identify and prioritize critical CRs, thereby facilitating targeted operational improvements. Lastly, we use a case study incorporating online hotel reviews from Ctrip.com to confirm the suggested approach's practical applicability. Comparative analysis demonstrates the effectiveness and superiority.

1. Introduction

With the increasing complexity of global e-commerce ecosystems, the development of robust and intelligent online platforms has become a strategic imperative. Among various sectors, online hotel booking services have gained substantial momentum, primarily due to their convenience and accessibility for consumers. The market paradigm has progressively shifted from enterprise-centric models to customer-driven frameworks [1], underscoring the critical need to accurately capture and respond to evolving customer

* Corresponding author at: School of Management Science and Engineering, Shandong University of Finance and Economics, Jinan, China.
E-mail addresses: peide.liu@gmail.com (P. Liu), shixinming@mail.sdufe.edu.cn (X. Shi), xycmarch@mail.sdufe.edu.cn (Y. Xu), dangran@mail.sdufe.edu.cn (R. Dang), yifanwu@mail.sdufe.edu.cn (Y. Wu).

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expectations. To remain competitive in this dynamic landscape, enterprises must deliver personalized service offerings that are tightly aligned with customer preferences. A key enabler of this goal is the systematic acquisition and transformation of customer demand information into actionable strategies for product and service optimization [2]. To address this challenge, we adopt the Quality Function Deployment (QFD) approach, which is a cross-functional planning methodology aimed at enhancing customer satisfaction through structured improvements in products, services, and processes [3]. QFD offers two principal advantages: (1) it supports data-driven decision-making during the transformation and decomposition of customer needs, and (2) it provides a consistent framework for tracking shifts in demand over time. These characteristics render QFD particularly relevant for the hotel industry, where service quality and customer satisfaction are vital competitive differentiators. At the core of the QFD methodology lies the House of Quality (HoQ), which explicitly maps customer requirements (CRs) to product characteristics (PCs). Online customer reviews offer a rich, real-time data source for identifying CRs, thereby serving as an effective input for the HoQ framework [4]. In this study, we integrate machine learning techniques with the Importance-Performance Analysis-Kano (IPA-KANO) model to classify CRs, and further enhance prioritization through an Interval-Valued Intuitionistic Fuzzy (IVIF)-HoQ matrix. The IVIF approach enables the modeling of membership uncertainty and multi-valued preference, addressing the limitations of traditional fuzzy sets. This facilitates a more nuanced prioritization of operational improvement alternatives by incorporating the vagueness inherent in user-generated content.

Compared to conventional data collection methods such as surveys and interviews, which are often resource-intensive, online reviews provide a cost-effective, real-time, and user-generated alternative for evaluating performance [5]. Previous research has explored the linkage between service attributes and customer satisfaction by leveraging such reviews [6,7]. As a result, the efficient extraction of actionable insights from online reviews has become increasingly vital for hotel operators. To this end, this study proposes a hybrid framework that combines machine learning with multi-criteria decision-making (MCDM) to enhance hotel operation management. Specifically, we employ the Latent Dirichlet Allocation (LDA) model, a well-established probabilistic topic modeling technique, to extract thematic structures from textual data. LDA facilitates the construction of topic-word and document-topic distributions, thereby supporting downstream tasks such as classification and clustering. For instance, Sutherland et al. successfully applied LDA to analyze online accommodation reviews, identifying distinctive service characteristics and competitive advantages [8].

However, a critical factor influencing the performance of LDA is the appropriate selection of the number of topics. This parameter significantly affects model accuracy, interpretability, and generalization capacity [9]. Standard techniques for determining the optimal number of topics include perplexity and coherence metrics [10,11]. Perplexity measures the model's predictive performance on unseen data but fails to capture semantic consistency. Coherence addresses this limitation by quantifying the interpretability and contextual alignment of top-ranked words within each topic. To balance statistical rigor and semantic clarity, we propose a comprehensive evaluation strategy that jointly considers perplexity and coherence scores. This strategy involves plotting both metrics against varying topic numbers to identify a region of optimal trade-off. Typically, perplexity decreases and coherence increases until saturation is reached. Selecting a topic count that balances these trends ensures robust topic modeling outcomes [12].

While LDA excels at identifying topics, it falls short in analyzing the sentiment orientation of textual content. Here, advancements in deep learning present a transformative opportunity. Neural network-based models, such as long short-term memory (LSTM) and bidirectional long short-term memory (BiLSTM), have demonstrated notable capabilities in processing sequential data [13]. However, the emergence of transformer-based models like bidirectional encoder representations from transformers (BERT) has redefined the state of the art in natural language processing (NLP), achieving unprecedented accuracy in tasks like sentiment analysis. The robustly optimized BERT pretraining approach (RoBERTa), an enhanced version of BERT, further advances these capabilities by improving the training methodology and data preprocessing pipeline. RoBERTa offers several advantages that make it particularly suitable for sentiment orientation analysis [14]. Firstly, RoBERTa removes BERT's next sentence prediction (NSP) task and focuses entirely on masked language modeling, resulting in richer contextual representations of textual data. Secondly, its bidirectional transformer architecture allows it to consider the context of a word from both preceding and succeeding text, which is crucial for accurately determining sentiment orientation in complex or ambiguous sentences. Thirdly, RoBERTa's robustness to diverse datasets makes it adaptable to domain-specific sentiment tasks, including customer reviews in the hospitality industry. By integrating LDA with RoBERTa, we propose a hybrid framework that extracts meaningful topics from online reviews and analyzes their sentiment orientation with high precision. This combination leverages LDA's strengths in topic modeling and RoBERTa's superior capabilities in sentiment analysis, enabling a comprehensive understanding of customer feedback [15]. Consequently, the proposed approach provides actionable insights into customer needs and preferences, supporting improved decision-making in service attribute allocation and product design. To accurately solve its weight problem, we attended that taking use of term frequency-inverse document frequency (TF-IDF) [16] technology and an IPA-combined Kano model (CKM) approach to adjust the weight of CRs Integrating the IPA and Kano models provides a comprehensive framework for categorizing service attributes. Combining IPA with Kano eliminates the one-dimensional focus of the Kano model, standardizing CR weights. Integrating machine learning with the Kano-IPA model supports optimal service element allocation based on online reviews.

An essential step in this study is prioritizing hotel operation improvement strategies under IVIF by integrating QFD with MCDM techniques. The QFD method is combined with IVIF to address uncertainties in online hotel reviews and reflect the varying evaluations of various decision-makers (DMs) [17]. Although earlier studies, such as those on online review-based group decision-making for evaluating passenger satisfaction [18], have offered foundational insights, they do not incorporate structured requirement transformation or multi-criteria prioritization under uncertainty. In contrast, this study proposes a hybrid IVIF-QFD-VIKOR model enhanced with machine learning-based sentiment analysis to support data-driven operational decisions tailored explicitly for the hospitality industry. The proposed IVIF-HOQ model provides a comprehensive framework for linking hotel operational features to customer needs. PCs and their weights are derived based on expert input and the IVIF-HOQ matrix. The MCDM model is then integrated with QFD to identify service elements and determine their priority levels. Additionally, the QFD method is combined with the

VlseKriterijumska Optimizacija I Kompromisna Resenje (VIKOR) method [19], enabling hotel managers to select compromise solutions by maximizing group utility and minimizing individual regret. The combined IVIF-QFD-VIKOR approach facilitates significant improvements in hotel operations and service quality. While some studies have combined QFD with MCDM techniques, such as DEMATEL [20] and TOPSIS [21], these do not consider machine learning-driven input extraction. Research on the integration of machine learning, QFD, and MCDM in the hotel management domain remains limited, thus underscoring the novelty and relevance of the present work.

Traditional weight methods often fail to capture comprehensive data. Currently, research data on the problem of extracting service attributes for categorization are mainly obtained through questionnaires. Nonetheless, acquiring data using questionnaires presents numerous disadvantages; for instance, it requires substantial time and financial resources, and the quality of the data is contingent upon the intricacy or length of the questionnaire, as well as the respondents' willingness to engage.

The conventional IPA-Kano approach suffers from several limitations that undermine the reliability of its resource allocation outcomes. Notably, its classification of Kano attributes often depends on subjective methods such as questionnaires, while overlooking sentiment analysis. Therefore, integrating machine learning-based sentiment analysis into the IPA-Kano model becomes essential.

The present text analysis, especially the machine learning model, mainly focused on English to train and improve. However, there are different languages worldwide, such as Chinese; therefore, processing and analyzing Chinese text data is challenging. In addition, the uncertainties of the evaluation of online comments are few to consider by scholars.

Few studies combine the QFD with MCDM to investigate enhancements in hotel operations by analyzing customer reviews from online platforms. Previous research has predominantly relied on hybrid methods for hotel selection, neglecting to incorporate CRs in identifying necessary improvements. While previous studies have explored hotel selection behaviors, there needs to be more understanding regarding how hoteliers can enact improvements based on CRs and PCs.

The key motivation behind our research design was to ensure a methodological synergistic effect, using each tool to complement the limitations of others and to improve the logical coherence of the framework. Therefore, our study introduces a QFD and VIKOR method model combined with IVIF to root in genuine CR insights to facilitate hotel operation enhancement. This paper incorporates the following contributions:

Utilize a crawler and the LDA topic model to identify CRs from online hotel platforms. A data-driven methodology gathers authentic online reviews to represent CRs and their preliminary objective weights. In contrast to conventional customer relationship determination procedures, this approach enables hoteliers to collect and utilize valuable information, enhance client satisfaction, and augment the hotel's operational competitiveness.

Determine CRs' weights by combining a deep learning model with a proposed IPA-CKM approach. Combined with the machine learning method, the IPA-CKM Model can eliminate questionnaire dependence. The data is more reliable and comprehensive than traditional CR determination methods. In addition, this method could improve product competitiveness and provide more accurate data to help hotel operations. RoBERTa, which is rarely used for studying online hotel reviews, could accurately capture and analyze textual sentiment and help enterprises understand the emotional trends of customer feedback.

Show the differences between various DMs and reflect the evaluation uncertainties. The ambiguity in CRs and PCs is addressed using the QFD and VIKOR approaches in conjunction with IVIF sets. This method can improve hotel sustainability and help businesses stand out in fiercely competitive markets.

Determine PCs' weight using the IVIF-QFD matrix and obtain the priorities of hotel investment alternatives based on a proposed IVIF-QFD-VIKOR method. By integrating QFD and MCDM, DMs can make more effective product designs and decisions by considering the requirements of multiple stakeholders and attributes. This combination enhances the transparency and consistency of decision-making and guarantees that the final product meets diverse market requirements.

To enhance the comprehensiveness and rationality of decision-making in service improvement, this study integrates four complementary methods: sentiment analysis with machine learning, the proposed combined Kano classification approach, and the VIKOR method with the IVIF-QFD model. Each method plays a distinct yet interconnected role in the framework, collectively bridging the gap between subjective customer feedback and objective managerial action. In conclusion, this integrated methodology enables a logically coherent and theory-grounded transformation from raw customer feedback to strategic recommendations. By combining behavioral modeling, fuzzy logic, and decision science, the proposed framework significantly enhances the rationality and explanatory power of service improvement decisions.

This paper is arranged as follows, with the context of the various sections displayed. Section 1 provides a review of the relevant literature. Section 2 summarizes the foundational concepts. Sections 3 and 4 detail the methodology developed in this study. Section 5 demonstrates the practicality of the proposed model through a case study analyzing online hotel reviews gathered from Ctrip.com. Section 6. offers a discussion that includes comparative and sensitivity analyses. Lastly, the conclusion and suggestions for future research are outlined in the final section.

2. Preliminaries

2.1. LDA model

LDA is a powerful approach for extracting service-related elements from online reviews [22]. It functions as an unsupervised probabilistic generative model capable of uncovering hidden topics within extensive collections of reviews. Within the framework of LDA, latent topics are conceptualized as coherent groupings of semantically related terms. Each review is assumed to be a probabilistic composition of multiple such topics, where a distinct probability distribution over the vocabulary characterizes each topic. The generative process entails sequentially drawing topics from the review-specific topic distribution and subsequently generating words based on the corresponding topic-word distribution. Following this procedure, the likelihood of each word appearing in a generated comment can be represented using a probabilistic formula [22]:

$$p(word|review) = \sum_{topic}(word|topic) \times p(topic|review) \tag{1}$$

Then, the probability formula is shown in the form of a matrix as follows: Fig. 1.

2.2. KANO model

Professor Noriaki Kano proposed the KANO model, inspired by the dual-factor theory of behavioral scientist Fredrick Herzberg [23]. The Kano model categorizes CRs into five distinct classes based on the correlation between various quality aspects and customer satisfaction; each attribute influences customers in unique ways [24]:

(1) Must-be quality (M): The product must have these qualities; otherwise, it will be useless and not create new customer satisfaction.

(2) Attractive quality (A): The customer will be satisfied if this quality can be realized. Meanwhile, customers will not be happy if this function is implemented.

(3) One-dimensional quality (O): Quality is positively correlated with customer satisfaction. Customer satisfaction increases with the function realization level and vice versa.

(4) Indifferent quality (I): This quality will not affect the product and customer satisfaction, whether it is presented or not.

(5) Reversed quality (R): Once this quality is realized, it will lead to customer dissatisfaction (Fig. 2).

2.3. IPA-KANO model

Kuo et al. [25] proposed a combined method of the IPA-Kano model, and this model could classify and analyze service attributes. To effectively characterize and distinguish the categories within the IPA-Kano model, familiar terminology is used for naming. Fig. 3 illustrates the specific categories and their corresponding scores as defined in the IPA-Kano model. The scores are ranked from 1 to 6, increasing from low to high [26].

2.4. IVIFS theory

This section presents core concepts of intuitionistic fuzzy sets (IFS) and interval-valued intuitionistic fuzzy sets (IVIFS).

Definition 1. ([27]) *Assume that $X = (x_1, x_2, ..., x_n)$ is a set. A fuzzy set A' in X is as follows:*

$$A' = \{\langle x, \mu_{A'}(x)|x \in X\rangle\ \} \tag{2}$$

where $\mu_{A'} : x \to [0, 1]$ is the membership function and $\mu_{A'}(x)$ is the membership degree of $x \in X$ to A'.

Definition 2. ([28]) *Assume that $X = (x_1, x_2, ..., x_n)$ is a set, and an IFS A is:*

$$A = \{\langle x, \mu_A(x), \nu_A(x)|x \in X\rangle\ \} \tag{3}$$

where the membership degree and nonmember-ship degree of x to A is $\mu_A(x)$ and $\nu_A(x)$. It satisfies for all $x \in X$, $0 \le \mu_A(x) +$

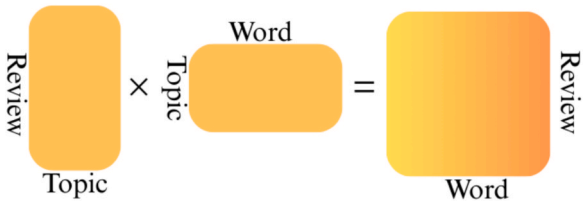


Fig. 1. The probability formula of LDA [22].

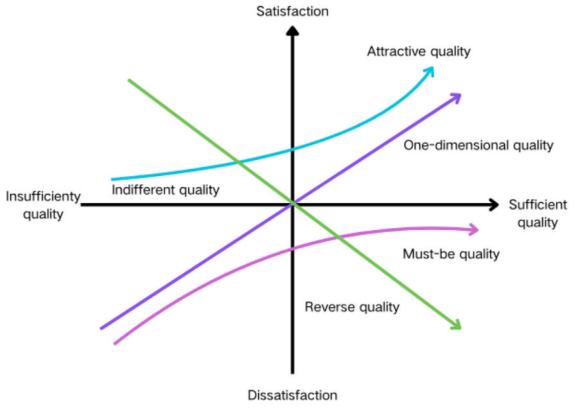


Fig. 2. Kano Model Diagram [24].

Strategic priorities	IPA-KANO categories	Importance	Performance
Keep up the good work (1)	Survival	High	High
Improvement (1)	Fatal	High	Low
Improvement (2)	Chronic disease	Low	Low
Keep up the good work (2)	Fitness	Low	High
Keep up the good work (3)	Major weapon	High	High
Improvement (3)	Defenseless strategic point	High	Low
Improvement (4)	Defenseless zone	Low	Low
Keep up the good work (4)	Supportive weapon	Low	High
Keep up the good work (5)	Precious treasure	High	High
Improvement (5)	Dusty diamond	High	Low
Improvement (6)	Rough stone	Low	Low
Keep up the good work (6)	Beginning jewellery	Low	High

Fig. 3. Relative categories and their scores of the IPA-Kano model [26].

$\nu_A(x) \leq 1$, and $\mu_A(x)$,$\nu_A(x) \in [0,1]$.

Definition 3. ([28]) *Let* X *be a nonempty set, and IVIFS* $\overline{A}$ *will be introduced as follows*:

$$\widetilde{A} = \left\{ \left\langle x, \widetilde{\mu}_{\widetilde{A}}(x), \widetilde{\nu}_{\widetilde{A}}(x) | x \in X \right\rangle \right\} \tag{4}$$

where the membership interval and nonmember-ship interval of $x \in X$ to $\widetilde{A}$ is $\widetilde{\mu}_{\widetilde{A}}(x) = \left[\mu_{\widetilde{A}}^L, \mu_{\widetilde{A}}^U\right] \subset [0,1]$ and $\widetilde{\nu}_{\widetilde{A}}(x) = \left[\nu_{\widetilde{A}}^L, \nu_{\widetilde{A}}^U\right] \subset [0,1]$. It satisfies for all $x \in X$, $\widetilde{\mu}_{\widetilde{A}}(x) + \max\widetilde{\nu}_{\widetilde{A}}(x) \leq 1$, and $\pi_{\widetilde{A}}(x) = \left[\pi_{\widetilde{A}}^L, \pi_{\widetilde{A}}^U\right]$ is the hesitation interval of x to A, where $\pi_{\widetilde{A}}^L = 1 - \mu_{\widetilde{A}}^U - \nu_{\widetilde{A}}^U, \pi_{\widetilde{A}}^U = 1 - \mu_{\widetilde{A}}^L - \nu_{\widetilde{A}}^L$.

Definition 4. ([29]) $\widetilde{a} = ([a,b],[c,d])$ *is an interval-valued intuitionistic fuzzy number (IVIFN), with the condition of* $0 \leq a \leq b \leq 1, 0 \leq c \leq d \leq 1$, *and* $b + d \leq 1$. *The score function* $S(\widetilde{a})$ *and the accuracy function* $A(\widetilde{a})$ *of are expressed as follows*:

$$S(\widetilde{a}) = \frac{a + b - c - d}{2}, \tag{5}$$

$$A(\widetilde{a}) = \frac{a + b + c + d}{2}. \tag{6}$$

Definition 5. ([30]) *Let* $\widetilde{a}_1 = ([a_1,b_1],[c_1,d_1])$ *and* $\widetilde{a}_2 = ([a_2,b_2],[c_2,d_2])$ *be two IVIFNS, and their comparison are as follows*:

(1) If $S\left(\widetilde{a}_1\right) > S\left(\widetilde{a}_2\right)$, then $\widetilde{a}_1 > \widetilde{a}_2$.

(2) If $S\left(\widetilde{a}_1\right) = S\left(\widetilde{a}_2\right)$, then.

(a) if $A\left(\widetilde{a}_1\right) > A\left(\widetilde{a}_2\right)$, then $\widetilde{a}_1 > \widetilde{a}_2$.

(b) if $A\left(\widetilde{a}_1\right) = A\left(\widetilde{a}_2\right)$, then $\widetilde{a}_1 = \widetilde{a}_2$.

Definition 6. ([31]) *Let* $\widetilde{a}_1 = ([a_1, b_1], [c_1, d_1])$ *and* $\widetilde{a}_2 = ([a_2, b_2], [c_2, d_2])$ *be two IVIFNS, and their Hamming distance is*:

$$D_H\left(\widetilde{a}_1, \widetilde{a}_2\right) = \frac{1}{4}(|a_1 - a_2| + |b_1 - b_2| + |c_1 - c_2| + |d_1 - d_2|). \tag{7}$$

Definition 7. ([32]) *Let* $\widetilde{a}_i = ([a_i, b_i], [c_i, d_i])(i = 1, 2, \ldots, n)$ *be IVIFNs and interval-valued intuitionistic fuzzy-weighted averaging (IIFWA) is the IVIF-weighted averaging operator.*

IIFWA: $R^n \to R$, where R is the set of IVIFNs, $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_n)^T$ is the weight vector of $\widetilde{a}_i$ with $\sum_{i=1}^{n} \lambda_i = 1$. So, aggregated IVIFN value is expressed as follows:

$$IIFWA_\lambda\left(\widetilde{a}_1, \widetilde{a}_2, \ldots, \widetilde{a}_n\right) = \lambda_1\widetilde{a}_1 \oplus \lambda_2\widetilde{a}_2 \oplus \ldots \oplus \lambda_n\widetilde{a}_n. \tag{8}$$

The Eq. (3) could be transformed into the following formula:

$$IIFWA_\lambda\left(\widetilde{a}_1, \widetilde{a}_2, \ldots, \widetilde{a}_n\right) = \left(\left[1 - \prod_{i=1}^{n}(1 - a_i)^{\lambda_i}, 1 - \prod_{i=1}^{n}(1 - b_i)^{\lambda_i}\right], \left[\prod_{i=1}^{n}(c_i)^{\lambda_i}, \prod_{i=1}^{n}(d_i)^{\lambda_i}\right]\right) \tag{9}$$

3. Data collection and processing

3.1. Data collection

This paper uses a crawler to get data on online hotel reviews to determine CR. Dr Tan Songbo collected a certain amount of online hotel review data from Ctrip.com and organized and classified it to form a corpus containing 10,000 pieces of data [33]. The choice of Ctrip.com is justified for several reasons: Firstly, the platform boasts a large user base and a wealth of user-generated reviews, providing an ample dataset for analysis. Secondly, Ctrip.com enforces strict rules regarding the publication of reviews, which contributes to the overall quality and reliability of the data.

Since 10,000 review data points from this corpus and 10,000 pure review data points are pre-cleaned, de-weighted, and then studied to make the final results more refined and accurate. The corpus of hotel online reviews compiled by Dr. Songbo Tan has some significant features, such as a large quantity, high quality, and strong usability, so it is used frequently. Therefore, we chose the corpus of hotel online reviews compiled by Dr. Songbo Tan for analysis.

Fig. 4 shows an example of an online review on Ctrip.com. It consists of five main sections:

(1) User information

(2) User reservation information

(3) Relevant scores of evaluations.

(4) Comments and related pictures.



Fig. 4. An Example of Online Reviews.

(5) The time and place of publication

3.2. Data preprocessing & processing

To optimize the extraction of service attributes, we preprocess the online reviews. These reviews contain both relevant terms related to service and attributes, as well as a significant amount of unrelated content. Once the target reviews are collected, data preprocessing and the selection of suitable evaluation criteria are essential for constructing the decision matrix. This preprocessing process consists of five key steps: (1) word segmentation, (2) part-of-speech tagging, (3) lemmatization, (4) stemming, and (5) stop-word removal. Initially, each review is segmented into individual words, followed by part-of-speech tagging to assign the correct grammatical role to each word. Then, remove the words unrelated to the hotel services, although they appear frequently. In addition, TF-IDF is used for word vector conversion. The above process can convert the text data into a language that the computer and the model can recognize.

We used the Jieba library, one of the most widely adopted tools for Chinese word segmentation. Specifically, we employed Jieba's "precise mode" to segment Chinese review texts into accurate token units. However, standard Jieba dictionaries may not recognize domain-specific terms relevant to the hotel industry or sentiment analysis. To enhance domain coverage and segmentation accuracy, we developed a customized lexicon, which was integrated into Jieba before segmentation. The custom lexicon was compiled through a two-pronged approach to ensure both comprehensiveness and objectivity:

a) Sentiment Analysis Vocabulary (Data-Driven Selection):

We first identified high-frequency sentiment-bearing terms from the raw corpus of 10,000 hotel reviews. These terms were filtered for relevance to hotel service experiences (e.g., comfort, noise, cleanliness, attitude), manually screened by three independent annotators, and cross-referenced against commonly used sentiment lexicons in Chinese NLP literature. Only those terms that appeared frequently and were verified for contextual relevance were retained.

b) Domain-Specific Vocabulary (External Source Integration):

We incorporated specialized terms from the Sogou Cell Dictionary, focusing on hotel-specific expressions such as hotel names, room types, service categories, and location markers. The Sogou Cell Dictionary is a publicly available and widely used linguistic resource in Chinese NLP applications, which supports generalizability and transparency.

All terms from both sources were added to Jieba using its standard *add_word()* and *load_userdict()* functions. This ensured seamless integration and improved recognition of critical terms during segmentation. Furthermore, we constructed a comprehensive stop-word list by aggregating and deduplicating entries from Baidu, Harvard, Sichuan University, and other reputable sources to enhance linguistic noise reduction. To further support reproducibility and transparency, we now provide a representative subset of the custom lexicon in the Appendix (Appendix 1).

After completing Chinese word segmentation, we need to refer to the deactivation table to delete all the deactivated words in the word segmentation. In this paper, we summarize the common deactivated words into an extensive deactivated word list, taking all the cases into account to provide higher-quality data for the subsequent work. The deactivation lists used here are the Baidu deactivation list, the Chinese deactivation list, the Harvard deactivation list, the Sichuan University Machine Intelligence Laboratory deactivation list, and the complete Chinese deactivation list, which is summarized, de-weighted, and screened. After the above two critical steps are completed, we can use the LDA topic model for modeling.

4. Methodology

The study incorporates three stages, as illustrated in Fig. 5. Firstly, the weights of CR criteria in QFD are ascertained by the IPA-CKM model utilizing data-driven methods. Thereafter, the relationships between CRs and PCs within the QFD matrix are analyzed to compute the weights of PCs by integrating them with the respective CR weights. These derived weights are then utilized in the third

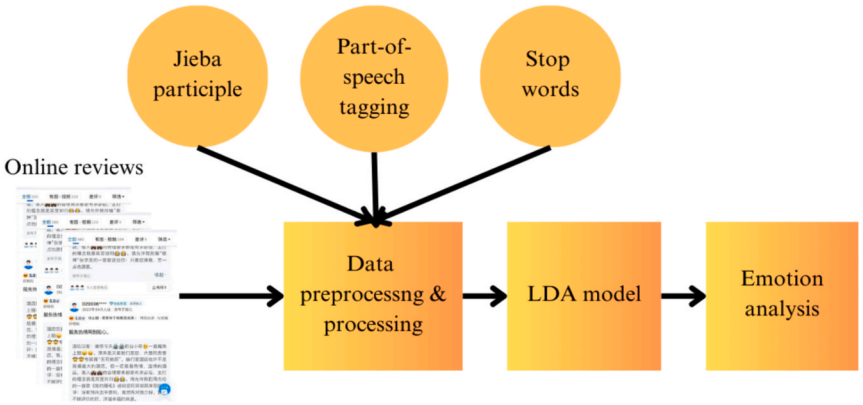


Fig. 5. Data pre-processing model.

phase, where an IVIF number-based VIKOR approach is applied to rank and select the most appropriate improvement alternatives. Fig. 6 illustrates the main techniques used throughout the article to analyze CRs. According to the main steps of obtaining CRs, Fig. 6 clearly shows the specific methods employed at each stage.

4.1. Extraction of CRs based on the LDA model and RoBERTa

Firstly, we extract CRs through a data-driven method of Python, an LDA model, and a RoBERTa sentiment analysis model. The specific steps are shown in Fig. 7 as follows.

Since the online reviews from the related website belong to non-structural context information, online reviews cannot be used to do the relative decision-making analysis. To analyze the structural data using the IPA-CKM model, we convert the online reviews into the structural data and explore positive and negative emotional tendencies towards service attributes. In this part, mining useful information based on online reviews mainly consists of two phases: (1) extraction of service attributes by the LDA model and (2) identification of sentiment orientation of service attributes based on RoBERTa.

4.1.1. Extraction of service attributes by LDA model

There are three steps in stage 1, and we introduce them in detail as follows:

(1) Determine the number of LDA topics that typically include the following methods.

Calculate Coherence: Consistency measures the quality of an LDA topic model. It determines a topic's consistency by evaluating the distribution of words in it. Calculate the coherence value for each topic and decide how many subjects have the highest coherence value. Topic coherence scores typically fall within the range [0, 1], where a score closer to 1 indicates stronger semantic consistency among the topic's keywords.

Evaluating Perplexity: Perplexity measures the ability to predict the test data. By determining the perplexity value of various topics, the number of topics with the lowest perplexity value is chosen, as a lower perplexity value typically indicates that the model is more effective in fitting the data.

Elbow Method: The Elbow Method is an empirical approach that examines a graph of Perplexity or Coherence values concerning the topics' amount, identifying the optimal one at the point when the curve exhibits an "elbow" or begins to flatten.

Combining these three steps can help determine the most appropriate number of topics. By calculating and comparing the Coherence and Perplexity values for different numbers of topics and by looking at the elbow curve, we can find a balance between a high degree of consistency and a low degree of perplexity in the model.

(2) Train the LDA model using Python to obtain the best topics and their related matrix.

Train the LDA model based on the generative process of LDA, and then obtain $X_{M\times N}$. There are two different matrixes will be obtained, which are the "Review-Topic" and "Topic-Word" Matrixes, as well as a topic list. $R = \{r_1, r_2, ..., r_M\}$ represents the set of online reviews, where r_m represents the mth online review, and M denotes the number of online reviews $(m = 1, 2, ..., M)$. Different topics may have similar meanings, then combine them, and choose the related topics to name them. Every topic can be thought of as a service attribute. So, I represents the extracted topic, and J_i represents the number of high-frequency words of the ith topic. The ith topic is defined as $f_i = \{word_{i1}, word_{i2}, ..., word_{ij}\}$, which means the jth frequency word of the ith topic, $i = 1, 2, ..., I, j = 1, 2, ...J_i$.

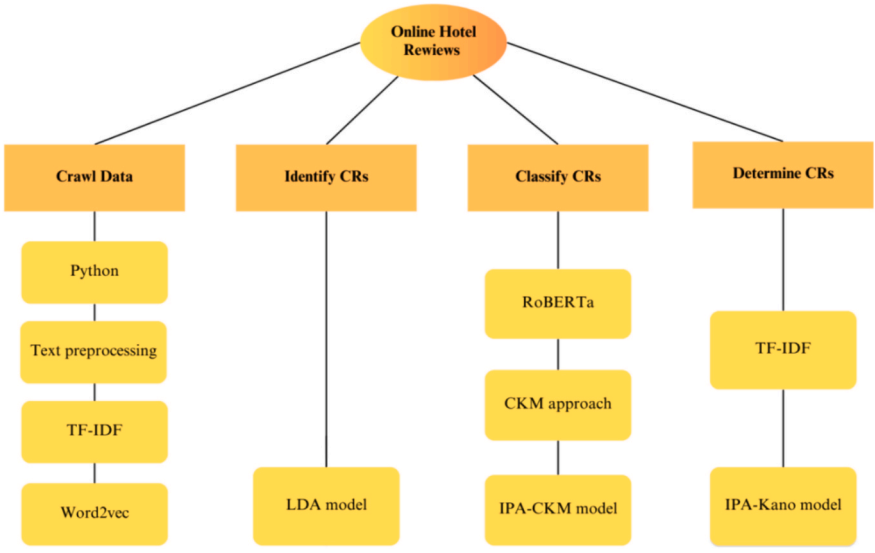


Fig. 6. The specific methods of analyzing CRs.

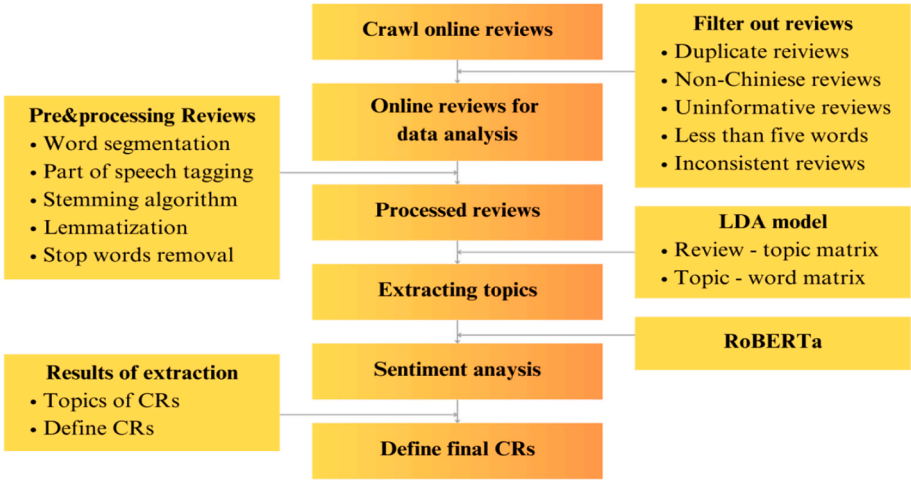


Fig. 7. Specific processes of extracting CRs.

4.1.2. *Identifying sentiment orientation based on RoBERTa model*

By enhancing text encoding, switching from static to dynamic masks, and training with bigger batches, RoBERTa is optimized using the BERT model [15]. First, we set up a test set to label the content of online reviews under different topics. Then, we used the test set to test the real online reviews by RoBERTa in Python. Finally, we obtain the result of sentiment orientation analysis based on RoBERTa. Based on the sentiment orientation result of every online review r_M, it is possible to determine the nominally coded data about the topics $f_i, i = 1,2,...,I, m = 1,2,...,M$, which is shown in Table 1.

Let * represent the sentiment orientation of online review r_m about attribute f_i, in which $* \in \{Pos, Neg\}, i = 1,2,...,I, m = 1,2,...,M$. Nominal encoded data is converted to structured data and the result is shown in Table 2. According to the rule that online review r_m about attribute f_i is positive orientation of sentiment, so $S_m^{Pos} = 1$, and $S_m^{\text{Neg}} = 0$;the online review r_m about attribute f_i is negative orientation of sentiment, so $S_m^{Pos} = 0$, and $S_m^{\text{Neg}} = 1, i = 1,2,...,I, m = 1,2,...,M$. The Structure data of online reviews are shown in Table 2.

$$S_m^* = \begin{cases} 1, & \text{if the sentiment orientation is } * \\ 0, & otherwise \end{cases}, i = 1,2,...,I, m = 1,2,...,M \tag{10}$$

According to the analysis of each review above, the number of sentiment orientation scores in every topic f_i could be calculated in Eq. (11) and Eq. (12).

$$S_{f_i}^P = \sum_{m=1}^{M} S_m^{\text{Pos}} \tag{11}$$

$$S_{f_i}^N = \sum_{m=1}^{M} S_m^{\text{Neg}} \tag{12}$$

4.2. *Classification of CRs based on the IPA-CKM model*

4.2.1. *The proposed CKM approach*

Integrating machine learning techniques with sentiment analysis and the KANO model offers a novel approach to capturing consumer preferences and enhancing strategies for product innovation and customer experience optimization. Each of these three methodologies brings unique strengths to the table. Machine learning provides powerful predictive tools, sentiment analysis extracts valuable insights from text data, and KANO modeling offers a structured framework for analyzing customer satisfaction. The key

Table 1
Nominally coded data of online reviews.

Online reviews	Attributes f_i			
	f_1	f_2	…	f_I
r_1	Pos	Pos	…	Neg
r_2	Neg	Pos	…	Pos
⋮	⋮	⋮	…	⋮
r_M	Pos	Pos	…	Neg

Table 2
Structure data of online reviews.

Online reviews	Attributes f_i						
	f_1		f_2		…	f_I	
	S_1^{Pos}	S_1^{Neg}	S_2^{Pos}	S_2^{Neg}	…	S_I^{Pos}	S_I^{Neg}
r_1	1	0	0	1	…	1	0
r_2	0	1	1	0	…	0	1
⋮	⋮	⋮	⋮	⋮	…	⋮	⋮
r_M	1	0	0	1	…	1	0

Note: *Pos* represents positive sentiment orientation, and *Neg* represents negative sentiment orientation in Tables 1 and 2.

innovation is creating a dynamic, data-driven approach to product development, customer satisfaction management, and feature prioritization. By integrating them, we can make a more nuanced, dynamic model that predicts and enhances customer satisfaction with a level of sophistication previously challenging to achieve. According to the RoBERTa sentiment analysis and the Kano Model's fundamental theory, we propose using CKM to identify the category of CRs. The proposed CKM is shown in Fig. 8. Unlike supervised learning approaches that rely on labeled training data, the Kano model is a conceptual framework derived from behavioral theory. Consequently, there is no universally accepted ground truth for categorizing CRs into Kano quality types, especially in the context of user-generated online reviews. To address this challenge and reduce subjectivity, we developed a rule-based hybrid classification strategy grounded in canonical Kano principles. This method integrates sentiment polarity, linguistic cues, and topic context to map CRs into the Kano framework.

To avoid the limitations of overly simplistic sentiment-based classification, we incorporated contextual review analysis. While sentiment orientation served as an initial filter, we further examined the linguistic and situational context of expressed dissatisfaction. For instance, although negative sentiment generally correlates with Must-be (M) quality attributes, in some cases, such as sound insulation, it requires deeper interpretation. Many user reviews revealed an implicit expectation that hotel rooms should be quiet. Deviations from this assumption were perceived as service failures, irrespective of other positive experiences. Thus, sound insulation was categorized as a Must-be (M) quality, reflecting its role as a baseline requirement rather than a performance-enhancing factor. This contextualized classification process enhances methodological robustness by capturing implicit customer expectations and ensuring the interpretive accuracy of Kano categorization in real-world, user-generated content environments.

The theoretical foundation of this combined model lies in integrating multiple established theories: (1) Expectation Disconfirmation Theory: It posits that customer satisfaction is determined by the gap between expected and perceived performance. By using sentiment analysis to extract perceptions and machine learning to predict how these perceptions will evolve, companies can anticipate customer expectations and manage satisfaction more effectively. (2) Cognitive and Affective Foundations of Consumer Behavior: The model integrates cognitive and emotional customer feedback elements. While machine learning and KANO modeling focus on the cognitive aspects of customer satisfaction, sentiment analysis adds a layer of affective insight, thus providing a more comprehensive understanding of customer needs. (3) Value Perception and Utility Theory: Machine learning models often rely on utility theory to

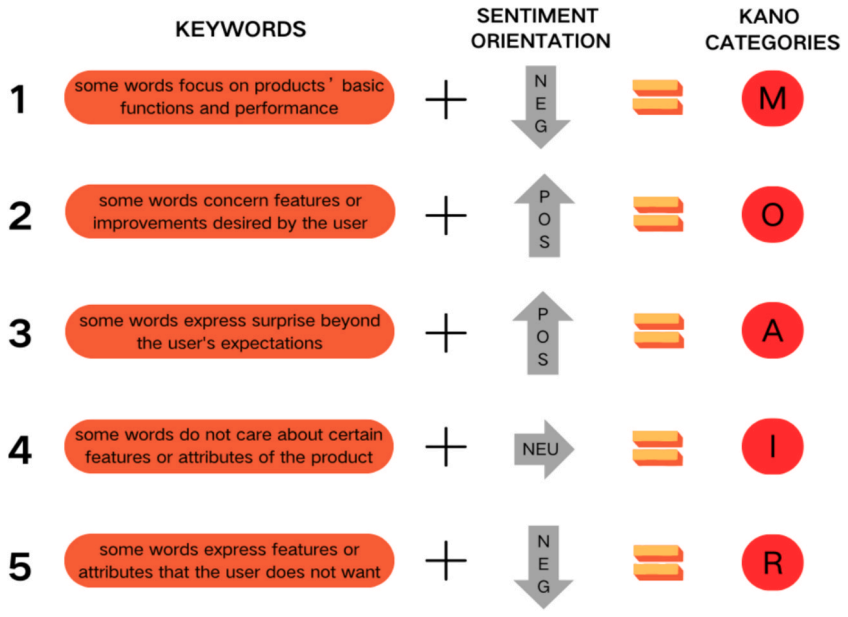


Fig. 8. CKM Approach.

predict the value customers place on different attributes. It is possible to prioritize features by combining the KANO model, which takes into account both their perceived value and their impact on total satisfaction. (4) Customer Satisfaction and Loyalty Models: Its integration with sentiment analysis and KANO categorization ties into broader customer loyalty, retention, and satisfaction theories. The model supports long-term customer satisfaction and brand loyalty by aligning product features with the factors that matter most to customers. In addition, there are some principles and mechanisms of the proposed method, as follows:

Data-Driven Insights. Sentiment analysis can extract relevant emotional insights from large amounts of feedback data, such as customer reviews and surveys, by utilizing machine learning to process them. These insights are then mapped to KANO categories. The sentiment data feeds into the KANO model, allowing a real-time understanding of customer satisfaction.
Customer Segmentation and Personalization. Machine learning can cluster customers based on behavior, while sentiment analysis helps to capture emotional nuances for each group. This segmentation informs how different customer types might perceive certain product features.
Predictive Modeling. Machine learning models can be trained to predict the effect of introducing or improving specific features. By combining sentiment analysis with KANO modeling, it is possible to estimate how changes to different product features might impact overall customer satisfaction and the likelihood of customer delight or dissatisfaction.
Dynamic Feature Evaluation. The traditional KANO model classifies features using surveys or interviews. However, using machine learning, the integrated model can continuously evaluate the relevance of features based on live customer feedback. This allows for real-time adaptations and improvements as customer needs and expectations evolve.

In conclusion, based on the above analysis, the LDA analysis, and the sentiment orientation analysis, a comprehensive judgment is made to categorize the reviews as one of the five types of needs in the Kano model:

(1) If the keywords in the reviews mainly focus on the product's basic functions or performance, and the affective orientation is negative, the reviews may be categorized as M.

(2) If the keywords in the review concern features or improvements the user desires, and the affective orientation is positive, it may be categorized as O.

(3) If the keywords in the review express surprise or pleasure beyond the user's expectations and the affective orientation is positive, it may be categorized as A.

(4) If the keywords in the review do not care about certain product features or attributes and the affective orientation is neutral, it may be categorized as I.

(5) If the keywords in the review express features or attributes that the user does not want and the affective orientation is negative, it may be categorized as R.

We conducted a consistency measurement analysis to assess the reliability and internal consistency of the sentiment-based Kano classification. The specific analysis will be shown in Appendix 2.

4.2.2. The new IPA-Kano model

The combined IPA-Kano model provides valuable guidance for managers, helping them understand how customers perceive different service attributes and develop effective strategies accordingly. This model categorizes service attributes and identifies which ones require prioritization or should remain at their current performance levels. Since online reviews often highlight factors that prevent negative experiences while still fulfilling customer needs, the IPA-Kano model in this study focuses on three primary types of attributes: one-dimensional, must-be, and attractive models. Each attribute is evaluated across two dimensions—importance (high or low) and performance (high or low). By utilizing the new IPA-Kano model, hotel managers will gain a detailed understanding of customer expectations and tailor their strategies for service enhancement.

First, the importance and performance of the 12 topics should be determined based on the IPA model, so $I = 12$. According to Table 2, the formula of the performance score for hotel attribute $f_i (i = 1, 2, ..., I)$ is shown as follows:

$$\mathrm{per}(f_i) = \frac{S_{f_i}^P}{S_{f_i}^P + S_{f_i}^N} \tag{13}$$

The IF-IDF model can determine the significance of each word in a document, and similarly, it helps assess the relevance of each attribute to overall customer satisfaction. The importance of each attribute is determined based on these calculations, applying the IF-IDF results derived from online reviews.

$$F_i = \sum_{i=1}^{I} F_{ij} (j = 1, 2, ..., J) \tag{14}$$

where F_i represents the result of IF-IDF for each review,

$$\mathrm{Imp}\ (f_i) = \frac{F_i}{\sum_{i=1}^{I} F_i} \tag{15}$$

This study integrates the CKM approach with the IPA-Kano model to thoroughly examine attribute characteristics and enhance the analysis of customer demands in the economic hotel sector. Equations (16) and (17) calculate the mean values for both importance and

performance.

$$Mean(Per) = \frac{S_{f_i}^{p} + S_{f_i}^{N}}{I} \tag{16}$$

$$\text{Mean (Imp)} = \frac{\sum_{i=1}^{I} F_i}{I} \tag{17}$$

If the score of performance and importance is more than the mean value, it would be high, and vice versa. In addition, according to Section 2.3, we categorize each topic into different series and find their final strategic priorities. The final weights of CRs are defined as follows.

$$\omega_i = \frac{h_i}{\sum_{i=1}^{I} h_i} \tag{18}$$

where $\omega_i (i = 1, 2, ..., I)$ present the final weights of CRs, and $h_i (i = 1, 2..., I)$ the represent the strategic priorities.

4.3. The IVIF-QFD method with MCDM for hotels

The QFD method is a well-established approach for translating CRs into product design specifications by using an HOQ matrix. This quantitative method is essential for PC optimization and for quickly meeting client needs. The IVIF-QFD method proposed here offers a structured framework to link customer needs with hotel service development. It helps in improving product design by converting the significance of CRs into the importance of PCs, supporting hotels in understanding customer needs, and guiding the development of new services and products.

To enhance hotel operations, this paper suggests the use of the IVIF-QFD-VIKOR method, which follows three main steps: 1) Determining the normalized weights of CRs through the IPA-Kano model; 2) Calculating the weights of PCs in the IVIF-QFD framework; and 3) Ranking alternative solutions using the IVIF-VIKOR approach.

Assuming that there are m CRs $\{CR_1, CR_2, ..., CR_m\}$ and n PCs $\{PC_1, PC_2, ..., PC_n\}$ in a QFD problem, then we proposed a CRs-PCs relation matrix to evaluate the relationship between m CRs and n PCs. Extract m CRs using online comments on the product with the same positioning, and identify n PCs through the enterprise's future technology research direction.

Firstly, we have obtained the importance and performance of all topics according to Eq. (13) and Eq. (15). The weight of CRs is calculated based on the result of importance divided by performance as follows.

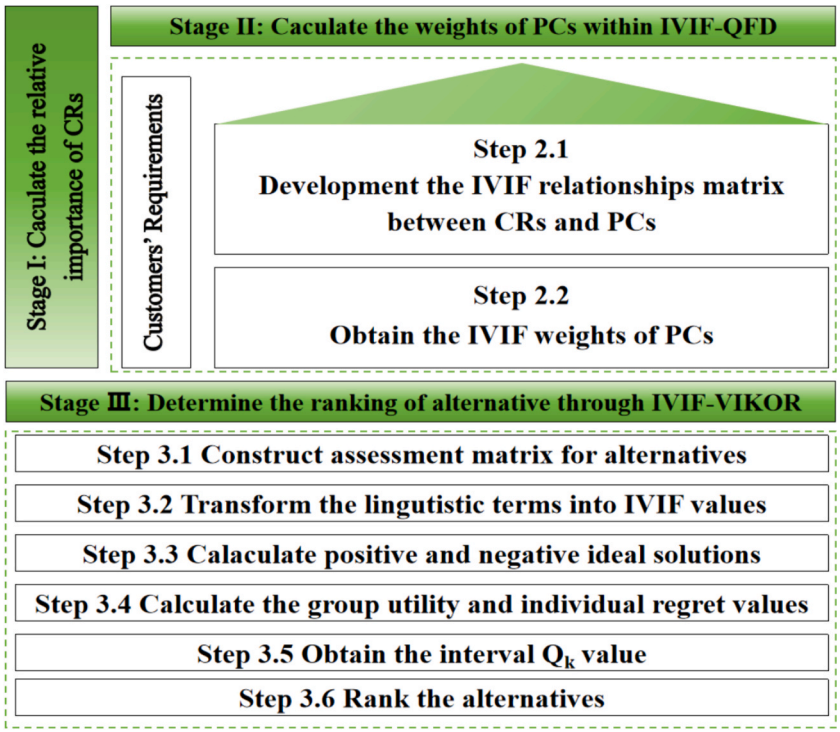


Fig. 9. Framework of the proposed methodology.

$$w_i = \frac{\mathrm{per}(f_i)}{\mathrm{imp}(f_i)} \tag{19}$$

Secondly, to ensure that the relative importance of different levels is normalized, each integer level $p, k \in \{1, 2, 3, 4, 5, 6\}$ is assigned a weight proportional to its value. We adjust the objective weights of CRs combined with the Kano model classification based on the priority of the IPA-Kano model as follows.

$$w_i^* = w_i \times \frac{p}{S}, S = \sum_{k=1}^{6} k. \tag{20}$$

Fig. 9 presents the overall framework of the proposed integrated approach, which combines IVIF logic with the QFD method and the VIKOR method to support decision-making in hotel service improvement. The framework is structured into three main stages:

Stage 1: Determine the weights of CRs through the IPA-Kano model.

This stage involves gathering customer feedback to determine the importance weights of CRs, which serve as the foundation for subsequent prioritization.

Stage 2: Calculate the weights of PCs using the IVIF-QFD model.

This stage consists of two key steps, illustrating how PCs influence CRs and defining the relationships between them under IVIF. The following steps can be used to obtain the weights of the PCs.

Step 1: Construct an IVIF relationship matrix between CRs and PCs in the QFD framework. Experts provide their evaluations of the correlations between CRs and PCs using IVIFNs, as outlined in Fig. 10 [32].

Step 2: The IVIF weights of PCs are derived from the QFD matrix. The primary function of this matrix is to identify the most critical PCs for meeting the CRs. However, the QFD framework is usually converted to a common way in a matrix called the HOQ. The six specific contexts of HOQ are shown in Fig. 11.

The relationship matrix and the weights of CRs are then integrated to ascertain the weights of PCs. The IVIF weights of PCs are calculated by the IVIF weight of CRs (w_j^*) multiplies by the IVIF relationships ($\widetilde{R}_J$) through Eq. (21) [34]

$$TW_i = \sum_j w_j^* \widetilde{R}_{Ji}, j = 1, 2, \ldots, n, i = 1, 2, \ldots, m. \tag{21}$$

where $\widetilde{R}_{Ji}$ shows the fuzzy relationship matrix between CRs and PCs; $\widetilde{\omega}_j$ indicates the weight of CRs; TW_i represents the total weight of PCs.

Then, the normalized IVIF weight of PCs that we could obtain is calculated in Eq. (22) [34].

$$N(TW_i) = \frac{S(TW_i)}{\sum_{i=1}^{m} S(TW_i)}. \tag{22}$$

Stage 3: Utilize IVIF-VIKOR model to determine the ranking of alternatives.

We proposed an improved VIKOR method combined with IVIF and determined the ranking of PCs that needed to improve through the IVIF-VIKOR model [35].

Step 1: Develop an evaluation matrix for the alternatives. The quantitative criteria are expressed through interval values, while the qualitative criteria are represented using linguistic terms, as illustrated in Fig. 10. A hybrid evaluation of these alternatives is subsequently derived.

Linguistic variables	Membership and non-membership degrees
Absolutely Low (AL)	([0,0.2], [0.5,0.8])
Very Low (VL)	([0.1,0.3], [0.4,0.7])
Low (L)	([0.2,0.4], [0.3,0.6])
Medium Low (ML)	([0.3,0.5], [0.2,0.5])
Approximately equal (AE)	([0.4,0.6], [0.2,0.4])
Medium High (MH)	([0.5,0.7], [0.1,0.3])
High (H)	([0.6,0.8], [0,0.2])
Very High (VH)	([0.7,0.9], [0,0.1])
Absolutely high (AH)	([0.8,1.0], [0,0])

Fig. 10. Linguistic scale and corresponding IVIFNs [32].

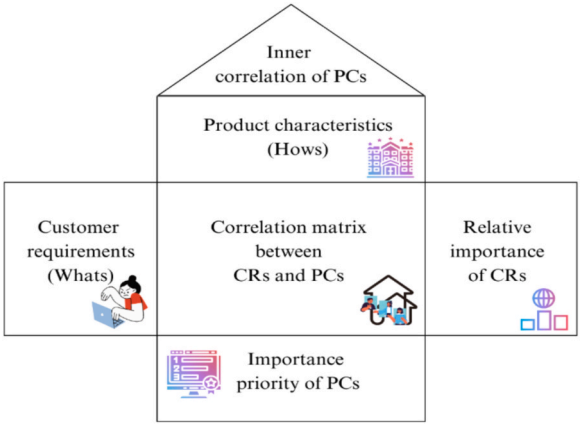


Fig. 11. HOQ model.

Step 2: Transform the linguistic terms into IVIF values to generate a group decision matrix $\widetilde{X}$, then $\widetilde{X}$ is defined as:

$$\widetilde{X}_{ki} = \left[\mu_{\widetilde{A}}^{L}(\widetilde{x}_{ki}), \mu_{\widetilde{A}}^{U}(\widetilde{x}_{ki})\right], \left[\nu_{\widetilde{A}}^{L}(\widetilde{x}_{ki})\nu_{\widetilde{A}}^{U}(\widetilde{x}_{ki})\right], \tag{23}$$

where $k = 1, 2, \ldots, n$ and $i = 1, 2, \ldots, m$. Interval values f_{ki} represents the evolution of k_{th} alternative concerning criterion. In addition, they are not exact numbers. Thus, they are denoted by $f_{ki} \in \left[f_{ki}^{L}, f_{ki}^{U}\right]$.

Step 3: The calculations for the optimal and worst ideal solutions for IVIFNs are conducted as $\widetilde{f}_i^{*}$ and $\widetilde{f}_i^{-}$ in Eq. (24)-(25). In addition, for interval values, the optimal and worst ideal solutions are f^{*} and f^{-} in Eq. (26)-(27), respectively.

$$\widetilde{f}_i^{*} = \left[\mu_{\widetilde{A}}^{L}(\widetilde{x}_i^{*}), \mu_{\widetilde{A}}^{U}(\widetilde{x}_i^{*})\right], \left[\nu_{\widetilde{A}}^{L}(\widetilde{x}_i^{*}), \nu_{\widetilde{A}}^{U}(\widetilde{x}_i^{*})\right], \tag{24}$$

where $[\mu_{\widetilde{A}}^{L}(\widetilde{x}_i^{*}) = \max_k(\mu_{\widetilde{A}}^{L}(\widetilde{x}_{ki})), \mu_{\widetilde{A}}^{U}(\widetilde{x}_i^{*}) = \max_k(\mu_{\widetilde{A}}^{U}(\widetilde{x}_{ki}))]$,$[\nu_{\widetilde{A}}^{L}(\widetilde{x}_i^{*}) = \max_k(\nu_{\widetilde{A}}^{L}(\widetilde{x}_{ki})), \nu_{\widetilde{A}}^{U}(\widetilde{x}_i^{*}) = \max_k(\nu_{\widetilde{A}}^{U}(\widetilde{x}_{ki}))]$ and

$$\widetilde{f}_i^{-} = \left[\mu_{\widetilde{A}}^{L}(\widetilde{x}_i^{-}), \mu_{\widetilde{A}}^{U}(\widetilde{x}_i^{-})\right], \left[\nu_{\widetilde{A}}^{L}(\widetilde{x}_i^{-}), \nu_{\widetilde{A}}^{U}(\widetilde{x}_i^{-})\right], \tag{25}$$

where $[\mu_{\widetilde{A}}^{L}(\widetilde{x}_i^{-}) = \min_k(\mu_{\widetilde{A}}^{L}(\widetilde{x}_{ki})), \mu_{\widetilde{A}}^{U}(\widetilde{x}_i^{-}) = \min_k(\mu_{\widetilde{A}}^{U}(\widetilde{x}_{ki}))]$,$[\nu_{\widetilde{A}}^{L}(\widetilde{x}_i^{-}) = \min_k(\nu_{\widetilde{A}}^{L}(\widetilde{x}_{ki})), \nu_{\widetilde{A}}^{U}(\widetilde{x}_i^{-}) = \min_k(\nu_{\widetilde{A}}^{U}(\widetilde{x}_{ki}))]$ and

$$f^{*} = \{f_1^{*}., \ldots, f_m^{*}\} = \left\{\left(\max_k f_{ki}^{U} | i \in J\right) or \left(\max_k f_{ki}^{L} | i \in J\right)\right\} i = 1, 2, \ldots, m, f^{-} = \{f_1^{-}., \ldots, f_m^{-}\} = \left\{\left(\max_k f_{ki}^{L} | i \in J\right) or \left(\max_k f_{ki}^{U} | i \in J\right)\right\} i = 1, 2, \ldots, m, \tag{27}$$

Including that, I and J separately represent benefit and worst criterion.

Step 4: Determine the group utility and individual regret values for IVIFN. They are represented as $S(A_k)$ in Eq. (28) and $R(A_k)$ in Eq. (29), and for interval value are $\left[S_k^{L}, S_k^{U}\right]$ in Eq. (31)-(32) and $\left[R_k^{L}, R_k^{U}\right]$ in Eq. (33)-(34) [36], respectively.

$$S(A_K) = \sum_{i=1}^{m}\left[N(TW_i)\frac{d\left(\widetilde{f}_i^{*}, \widetilde{x}_{ki}\right)}{d\left(\widetilde{f}_i^{*}, \widetilde{f}_i^{-}\right)}\right], \tag{28}$$

$$R(A_K) = \max_i\left[N(TW_i)\frac{d\left(\widetilde{f}_i^{*}, \widetilde{x}_{ki}\right)}{d\left(\widetilde{f}_i^{*}, \widetilde{f}_i^{-}\right)}\right], \tag{29}$$

where

$$d\left(\widetilde{f}_i^*,\widetilde{x}_{ki}\right)=\frac{1}{4}\begin{pmatrix}\left|\mu_{\widetilde{A}}^L(\widetilde{x}_i^*)-\mu_{\widetilde{A}}^L(\widetilde{x}_{ki})\right|+\left|\mu_{\widetilde{A}}^U(\widetilde{x}_i^*)-\mu_{\widetilde{A}}^U(\widetilde{x}_{ki})\right|\\+\left|\nu_{\widetilde{A}}^L(\widetilde{x}_i^*)-\nu_{\widetilde{A}}^L(\widetilde{x}_{ki})\right|+\left|\nu_{\widetilde{A}}^U(\widetilde{x}_i^*)-\nu_{\widetilde{A}}^U(\widetilde{x}_{ki})\right|\\+\left|\pi_{\widetilde{A}}^L(\widetilde{x}_i^*)-\pi_{\widetilde{A}}^L(\widetilde{x}_{ki})\right|+\left|\pi_{\widetilde{A}}^U(\widetilde{x}_i^*)-\pi_{\widetilde{A}}^U(\widetilde{x}_{ki})\right|\end{pmatrix}$$

$$d\left(\widetilde{f}_i^*,\widetilde{f}_i^-\right)=\frac{1}{4}\begin{pmatrix}\left|\mu_{\widetilde{A}}^L(\widetilde{x}_i^*)-\mu_{\widetilde{A}}^L(\widetilde{x}_i)\right|+\left|\mu_{\widetilde{A}}^U(\widetilde{x}_i^*)-\mu_{\widetilde{A}}^U(\widetilde{x}_i)\right|\\+\left|\nu_{\widetilde{A}}^L(\widetilde{x}_i^*)-\nu_{\widetilde{A}}^L(\widetilde{x}_i)\right|+\left|\nu_{\widetilde{A}}^U(\widetilde{x}_i^*)-\nu_{\widetilde{A}}^U(\widetilde{x}_i)\right|\\+\left|\pi_{\widetilde{A}}^L(\widetilde{x}_i^*)-\pi_{\widetilde{A}}^L(\widetilde{x}_i)\right|+\left|\pi_{\widetilde{A}}^U(\widetilde{x}_i^*)-\pi_{\widetilde{A}}^U(\widetilde{x}_i)\right|\end{pmatrix}. \tag{30}$$

Computing $\left[S_k^L,S_k^U\right]$ and $\left[R_k^L,R_k^U\right]$ interval as follows:

$$S_k^L=\sum_{i\in I}N(TW_i)\left(\frac{f_i^*-f_{ki}^U}{f_i^*-f_i^-}\right)+\sum_{i\in J}N(TW_i)\left(\frac{f_{ki}^L-f_i^*}{f_i^--f_i^*}\right)k=1,\ldots,n, \tag{31}$$

$$S_k^U=\sum_{i\in I}N(TW_i)\left(\frac{f_i^*-f_{ki}^L}{f_i^*-f_i^-}\right)+\sum_{i\in J}N(TW_i)\left(\frac{f_{ki}^U-f_i^*}{f_i^--f_i^*}\right)k=1,\ldots,n, \tag{32}$$

$$R_k^L=\max\left\{N(TW_i)\left(\frac{f_i^*-f_{ki}^U}{f_i^*-f_i^-}\right)\middle|i\in I,N(TW_i)\left(\frac{f_{ki}^L-f_i^*}{f_i^--f_i^*}\right)|i\in J\right\}k=1,\ldots,n, \tag{33}$$

$$R_k^U=\max\left\{N(TW_i)\left(\frac{f_i^*-f_{ki}^L}{f_i^*-f_i^-}\right)\middle|i\in I,N(TW_i)\left(\frac{f_{ki}^U-f_i^*}{f_i^--f_i^*}\right)|i\in J\right\}k=1,\ldots,n, \tag{34}$$

where $N(TW_i)$ is the weight of the criterion C_i.

Step 5: $Q_{k=}\left[Q_k^L,Q_k^U\right];k=1,2,\ldots,n$ is circulated by Eq. (35) and Eq. (36):

$$Q_k^L=\nu\frac{\left(S_k^L-S^*\right)}{\left(S^--S^*\right)}+(1-\nu)\frac{\left(R_k^L-R^*\right)}{\left(R^--R^*\right)}, \tag{35}$$

$$Q_k^U=\nu\frac{\left(S_k^U-S^*\right)}{\left(S^--S^*\right)}+(1-\nu)\frac{\left(R_k^L-R^*\right)}{\left(R^--R^*\right)}. \tag{36}$$

where

$$\begin{aligned}S^*&=\min_k S_k^L,S^-=\max_k S_k^U,\\R^*&=\min_k R_k^L,R^-=\max_k R_k^U.\end{aligned} \tag{37}$$

Set $\nu=0.5$, and ν is the weight of the maximum group utility.

Step 6: Rank the alternatives. The minimum $Q_{k=}\left[Q_k^L,Q_k^U\right]$ of all alternatives should rank first, which is the compromise result. The optimal level of DM is the parameter $\eta(0<\eta\le 1)$, with rational DM $\eta=0.5$. In addition, the corresponding comparison rules are shown in Fig. 12.

Minimum interval number between $\left[a^L,a^U\right]$ and $\left[b^L,b^U\right]$	
When $a^U<b^L\Rightarrow$	$\left[a^L,a^U\right]<\left[b^L,b^U\right]$
When $a^L<b^L<b^U<a^U$, if $\eta(b^L-a^L)\ge(1-\eta)(a^U-b^U)\Rightarrow$	$\left[a^L,a^U\right]<\left[b^L,b^U\right]$
When $a^L<b^L<b^U<a^U$, if $\eta(b^L-a^L)<(1-\eta)(a^U-b^U)\Rightarrow$	$\left[b^L,b^U\right]<\left[a^L,a^U\right]$
When $a^L<b^L<a^U<b^U$, if $\eta(b^L-a^L)\ge(1-\eta)(b^U-a^U)\Rightarrow$	$\left[a^L,a^U\right]<\left[b^L,b^U\right]$
When $a^L<b^L<a^U<b^U$, if $\eta(b^L-a^L)<(1-\eta)(b^U-a^U)\Rightarrow$	$\left[b^L,b^U\right]<\left[a^L,a^U\right]$

Fig. 12. Comparison of interval numbers [36].

5. Case study

This section shows a real-life example of hotel operation. This paper uses Dr. Tansongbo's database of real online reviews for our analysis. Dr. Tan Songbo collected a certain amount of online hotel review data from Ctrip.com and sorted and classified it to form a corpus containing 10,000 pieces of data.

5.1. Case background

Ctrip.com is a famous website in China, and there are two main reasons why we chose the data from Ctrip.com as the analytic target.

Data richness and diversity: As an online travel service platform, Ctrip.com provides a large number of tourist attractions and hotel information. By crawling the user review data on Ctrip.com, we can obtain detailed information on hotel reviews, prices, and accommodation experiences, all of which are of great value to customers when choosing hotels.
Market analysis and competition strategy: By crawling data from Ctrip.com, we can analyze hotel hotness, price, evaluation, and other indicators, provide customers with reasonable travel suggestions, and provide hoteliers with market dynamics and competition strategies.

5.2. Implementation

This paper proposed a QFD-MCDM method for improving hotels to solve these problems. This paper applies the IPA-Kano Model with a data-driven method to analyze the hotels of Ctrip.com operation quantitatively. Then, using the QFD method, the CRs and their weights are taken as the inputs of the left wall of the hotel's HoQ. The correlation matrix between CRs-PCs indicators and the auto-correlation matrix of PCs indicators is taken as the input of the room and the roof, respectively, to obtain the importance of the PCs indicators. Then, the VIKOR-QFD method is established to optimize the hotel's operating quality.

The detailed flow of the proposed QFD method is shown below. In the rest of this section, we use hotels on Ctrip.com as an example to illustrate the implementation process of the proposed IVIF-QFD-VIKOR method, including three stages. Fig. 13 shows the specific progress of the new methodology.

Stage 1: Find CRs and determine their relative importance by the LDA model, RoBERTa, and the IPA-Kano model

Step 1: Crawl the online comments. Here, we choose Dr. Tansongbo's database of real online reviews for our analysis. Dr. Tan Songbo collected a certain amount of online hotel review data from Ctrip.com and sorted and classified it to form a corpus containing 10,000 reviews.

Step 2: Apply Jieba technology to do the word segmentation.

Step 3: Remove stop words from the dataset based on a predefined stop word list to enhance the efficiency of the data mining process. These words, such as "OK," "yes," "eh," and "la," offer minimal value to the text's meaning but appear frequently. Generally, modal particles, pronouns, prepositions, and conjunctions fall under this category.

Step 4: Identify the optimal number of topics. A comprehensive approach is employed, combining both perplexity and coherence scores to determine the ideal topic count. The procedure is as follows:

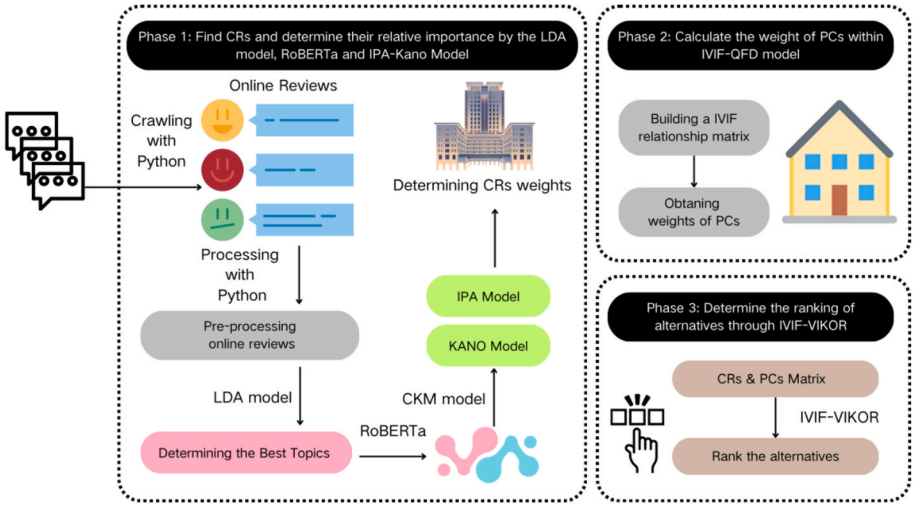


Fig. 13. The progress of the new Methodology.

Calculate Coherence: Compute the coherence value for each topic and select the number of topics with the highest coherence score. The perplexity result for 30 topics is presented in Fig. 14.
Evaluating Perplexity: Perplexity measures how well the model predicts unseen data. Using Python, we obtain the perplexity for 30 topics, as shown in Fig. 15.
The Elbow Method: This method involves analyzing the graph of perplexity or coherence values against the number of topics and selecting the point where the curve flattens, which represents the optimal number of topics.

These three steps work together to help determine the most suitable number of topics in the LDA model. By comparing both coherence and perplexity values and identifying the "elbow" in the graph, we can achieve a balance between model consistency and predictive ability. Coherence reaches its highest value (0.581) when the number of topics is 12, indicating that this number of topics provides the most semantically meaningful and interpretable topics. While complexity decreases as the number of topics increases, it begins to level off after 12 topics, indicating diminishing returns. This suggests that further increasing the number of topics does not bring substantial improvement, and the model may become unnecessarily complex. Thus, 12 topics represent a spot where there is an optimal trade-off between interpretability, which is ensured by coherence and predictive ability that is reflected by perplexity, while avoiding the risks of overfitting and unnecessary model complexity.

In addition, topic variance was employed as a complementary metric to evaluate the stability and discriminative capacity of the LDA model. A moderately low variance indicates that topics are neither overly sparse nor excessively generalized, supporting a well-balanced topic structure.

As illustrated in Appendix 3, across the 30 topic-number configurations, topic variance exhibits a generally decreasing trend (from 0.39 to 0.15). At 12 topics, the variance reaches 0.24, which falls within a distinctly stable zone, indicating a balanced distribution of topics across documents. This balance suggests that the model captures distinct latent structures while maintaining interpretability and robustness. Similar values have been reported in related studies as indicative of optimal model complexity [37,38]. This metric reinforces the model's robustness and helps mitigate risks of topic redundancy or overfitting, mainly when coherence and perplexity alone yield ambiguous optima.

Taken together, these results demonstrate that the 12-topic configuration offers a well-balanced trade-off among semantic clarity, model robustness, and complexity control. It avoids the generation of redundant or overly fragmented topics while preserving interpretability and generalizability.

In addition, by selecting 12 topics, we avoid introducing extraneous noise and unnecessary details that would complicate the model without adding valuable insights. Choosing 12 topics ensures a high degree of semantic consistency and keeps the computational cost manageable, making the model more practical for real-world applications. Therefore, after further evaluation, twelve topics are selected, and the analysis proceeds with Python in the LDA model.

Step 5: Apply the RoBERTa model to do the context analysis. The RoBERTa model was fine-tuned using Hugging Face's Training Arguments. We set the learning rate to 3e-5, the number of training epochs to 3, and the batch sizes to 32 (training) and 16 (evaluation). A linear learning rate scheduler with 500 warmup steps was used to stabilize early training. To mitigate overfitting, weight decay was set to 0.01. Evaluation was conducted at the end of each epoch, and mixed-precision training (fp16) was enabled to improve computational efficiency. The specific hyperparameter configuration for RoBERTa Fine-Tuning is shown in Table 3.

Step 6: Train the LDA model by Python to obtain the best topics and their related matrix. Train the LDA model and then obtain $X_{M\times N}$. Different topics may have similar meanings, then combine them, and choose the related topics to name them. Every topic can be thought of as a service attribute. So, I represents the extracted attribute (topic), and J_i represents the number of high-frequency words of the ith topic. The ith topic is defined as $f_i = \{word_{i1}, word_{i2}, \ldots, word_{ij}\}$, which means the j th frequency word of the i th topics,$i = 1,2,\ldots,I, j = 1,2,\ldots,J_i$. We obtain the LDA topics list, the number of comments, and coherence scores included in each topic in Table 4.

In addition, we obtain the results of the TF-IDF method for 12 topics by Python, which is shown in Table 5, which J_i represents the

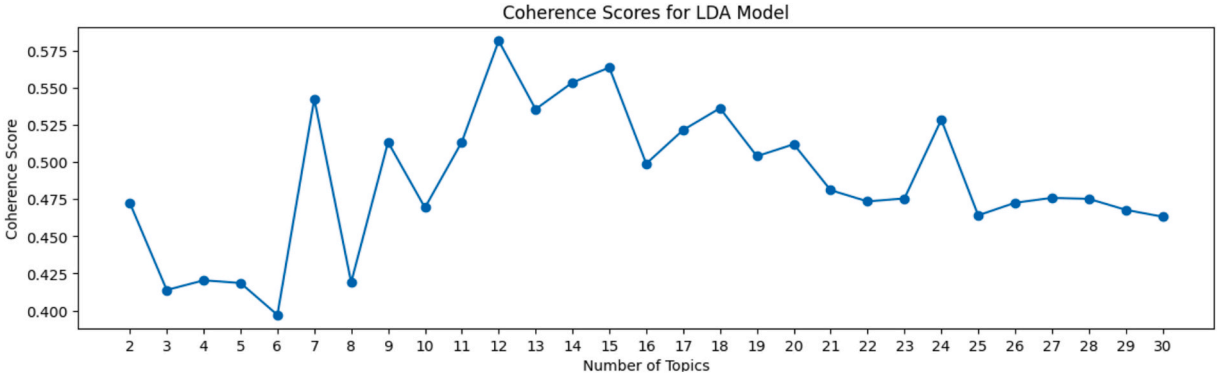


Fig. 14. Coherence Scores for the LDA model.

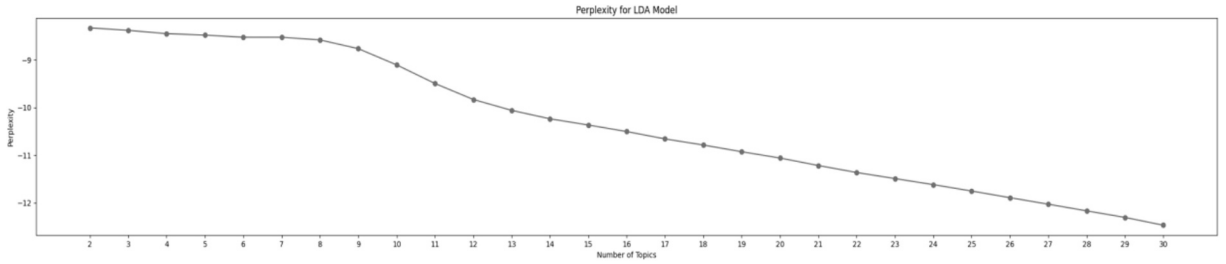


Fig. 15. Perplexity for LDA model.

Table 3
Hyperparameter Configuration for RoBERTa Fine-Tuning.

Hyperparameter	Description	Value
Learning Rate	Step size for updating model parameters; highly sensitive during fine-tuning	3e-5
Epochs	Number of complete passes through the training data	3
Training Batch Size	Number of samples per batch used during training	32
Evaluation Batch Size	Number of samples per batch used during validation	16
Warmup Steps	Gradual increase of learning rate at training start to stabilize convergence	500
Weight Decay	L2 regularization coefficient to prevent overfitting	0.01
Evaluation Strategy	Model is evaluated once at the end of each epoch	epoch
LR Scheduler Type	Linear learning rate scheduler with warmup followed by linear decay	linear
Mixed Precision (FP16)	Enabled to improve training speed and reduce memory usage	True

Table 4
Specific parameters related to the LDA Model.

No.	Topics	#	Coherence Scores
f_1	Acceptance of Opinions	91	0.712
f_2	Floor Characteristics	7	0.656
f_3	Way of Check-in	160	0.387
f_4	Hotel Environment	18	0.521
f_5	Comprehensive Evaluation	9243	0.659
f_6	Surroundings	24	0.656
f_7	Sound insulation	130	0.461
f_8	Service Characteristics	189	0.450
f_9	Service Experience	46	0.484
f_{10}	Room Types	38	0.719
f_{11}	Pick-up Service	17	0.545
f_{12}	Hotel Facilities	37	0.654

number of high-frequency words of the *i*th topic; F represents the frequency.

Therefore, we obtain each topic's importance in Table 6 based on Eqs. (24) and (25).

Step 6: Identify the sentiment orientation of topics by RoBERTa. We obtain the total sentiment score results in Table 7 based on Eq. (23). n represents the number of related comments in every topic, $S_{f_i}^P$ and $S_{f_i}^N$ represent the total positive and negative sentiment scores for every topic, respectively. $Per(f_I)$ represents the performance of topic f_i,$i = 1,2,...,I$.

Step 7: Combining the sentiment analysis results and the opinions of the three experts from the hospitality management program, we converted the subject lines to CRs $CR_i(i = 1,2,...,12)$. The specific CRs are shown in Table 8.

Step 8: According to the CKM approach, we classify all topics f_I into Kano Model in Table 9.

Analyze the keywords of all topics. According to the LDA model, we obtain the most frequent words of every topic and their frequency. We analyze the related keywords and study the specific reviews that have related keywords based on the proposed CKM approach. In Table 9, Numbers 1 to 5 represent different keywords.

Identify the sentiment orientation of all topics based on the $S_{f_i}^P$ and $S_{f_i}^N$.

If $S_{f_i}^P > S_{f_i}^N$, it will be a positive sentiment orientation;
If $S_{f_i}^P < S_{f_i}^N$, it will be a negative sentiment orientation;
If $S_{f_i}^P = S_{f_i}^N$, it will be a neutral sentiment orientation;

Table 5
The LDA Topics and their frequencies.

f_l	Topic Names	J_1	F	J_2	F	J_3	F	J_4	F	J_5	F
f_1	Acceptance of Opinions	反馈(Feedback)	0.037	期待(Expectation)	0.015	提出(Offering)	0.013	齐全(Complete)	0.011	支持 (Support)	0.011
f_2	Floor Characteristics	安静(Quiet)	0.022	偏高(High)	0.021	低(Low)	0.008	楼层(Floor)	0.007	窗户(Windows)	0.006
f_3	Way of Check-in	前台(Front Desk)	0.016	预定(Booking)	0.014	服务员(Waiter)	0.011	升级(Upgrade)	0.009	信用卡 (Credit Card)	0.008
f_4	Hotel Environment	适中(Moderate)	0.010	繁华(Prosperity)	0.009	优美(beautiful)	0.008	外部环境(Exterior)	0.007	接送 (Pick-up)	0.007
f_5	Comprehensive Evaluation	房间(Room)	0.071	服务(Service)	0.034	早餐(Breakfast)	0.023	入住(Check-in)	0.020	设施(Facilities)	0.015
f_6	Surroundings	海景(Sea View)	0.019	风景(Scenery)	0.013	印象(Impression)	0.013	优美(beautiful)	0.011	自助餐(Buffet)	0.011
f_7	Sound insulation	隔音(Soundproof)	0.021	声音(Sound)	0.020	空调(Air-condition)	0.017	隔壁(Neighbors)	0.014	效果 (Effect)	0.013
f_8	Service Characteristics	入住(Check-in)	0.032	前台(Front Desk)	0.020	退房(Check-out)	0.013	经理(Manager)	0.009	时间 (Time)	0.008
f_9	Service Experience	有礼貌(Courteous)	0.006	服务员(Waiter)	0.006	不足(Not good)	0.006	液晶电视(TV)	0.006	欣赏(Appreciation)	0.005
f_{10}	Room Types	套房(Suites)	0.011	很大(Large)	0.009	前台(Front Desk)	0.008	拖鞋(Slipper)	0.006	推荐(Recommendation)	0.006
f_{11}	Pick-up Service	机场(Airport)	0.023	接机(Pick-up)	0.015	站点(Site)	0.014	班车(Shuttle bus)	0.013	大巴(Bus)	0.012
f_{12}	Hotel Facilities	温泉(SPA)	0.021	健身房(Gym)	0.008	娱乐(Entertainment)	0.006	中规中矩(Medium)	0.006	早餐(Breakfast)	0.005
f_l	Topic Names	J_6	**F**	J_7	**F**	J_8	**F**	J_9	**F**	J_{10}	**F**
f_1	Acceptance of Opinions	更好(Better)	0.010	很好(Good)	0.009	宾客 (Guest)	0.009	有特色(Distinctive)	0.008	入住(Check-in)	0.008
f_2	Floor Characteristics	起码 (At least)	0.006	最贵(Expensive)	0.006	翻新(Refurbished)	0.005	害怕(Scared)	0.004	环境(Environment)	0.004
f_3	Way of Check-in	楼层(Floor)	0.008	结账(Check-out)	0.007	入住(Check-in)	0.007	退房(Check-out)	0.007	房间(Room)	0.007
f_4	Hotel Environment	星级 (Star)	0.005	酒楼(Restaurant)	0.005	整洁(Clean)	0.004	度假(Vacation)	0.004	积水(Cumulus)	0.004
f_5	Comprehensive Evaluation	很好(Good)	0.014	价格(Price)	0.013	环境(Environment)	0.013	干净(Clean)	0.011	前台(Front Desk)	0.010
f_6	Surroundings	周边环境(Surroundings)	0.011	套房(Suites)	0.010	海景房(Sea View)	0.010	方便的(Convenient)	0.010	晚餐(Lunch)	0.010
f_7	Sound insulation	房间(Room)	0.008	听到(Heard)	0.008	走廊(Hallway)	0.007	电话(Phone)	0.006	服务员(waiters)	0.005
f_8	Service Characteristics	员工(Staff)	0.008	电话(Phone)	0.007	情况(Situation)	0.007	预定(Order)	0.006	友好(Friendly)	0.006
f_9	Service Experience	友好(Friendly)	0.005	值班 (On Duty)	0.005	品尝(Tasting)	0.004	干净(Clean)	0.004	经理(Manager)	0.004
f_{10}	Room Types	客厅	0.006	高速(High-way)	0.006	卧室(Room)	0.005	物超所值	0.005	多样性(Diversity)	0.005
f_{11}	Pick-up Service	古城(Ancient City)	0.011	路过(Pass-by)	0.011	很漂亮 (Very nice)	0.010	导航(Navigation)	0.010	合作(Cooperation)	0.008
f_{12}	Hotel Facilities	洗衣(Laundry)	0.005	偏贵(Expensive)	0.005	海滩(Beach)	0.005	港币(HKD)	0.004	配有(Furnishing)	0.004

Note: In Chinese, the topic of Service Characteristics(“服务特性”) refers specifically to the inherent attributes and objective features of the service offering—such as speed, accuracy, or variety—whereas Service Experience (“服务体验”) captures the subjective and emotional responses of the customer when interacting with the service—such as satisfaction, comfort, or stress. These distinctions are well-recognized in the service management and customer experience literature in Chinese, and they are clearly differentiated in the annotated corpus used in our study.

Table 6
The LDA Topics and Their Importance.

f_I	Topic Names	$\sum F_I$	Imp(f_I)
f_1	Acceptance of Opinions	41.274	0.007
f_2	Floor Characteristics	3.235	0.001
f_3	Way of Check-in	83.017	0.013
f_4	Hotel Environment	9.116	0.001
f_5	Comprehensive Evaluation	5808.892	0.940
f_6	Surroundings	10.581	0.002
f_7	Sound insulation	66.0133	0.011
f_8	Service Characteristics	89.053	0.014
f_9	Service Experience	23.351	0.004
f_{10}	Room Types	20.131	0.003
f_{11}	Pick-up Service	7.879	0.001
f_{12}	Hotel Facilities	16.402	0.003

Table 7
The LDA Topics and Their Performance.

f_I	Topics	$S_{f_i}^P$	$S_{f_i}^N$	$Per(f_I)$
f_1	Acceptance of Opinions	48	43	0.527
f_2	Floor Characteristics	0	7	0.125
f_3	Way of Check-in	62	98	0.388
f_4	Hotel Environment	9	9	0.500
f_5	Comprehensive Evaluation	5069	4174	0.548
f_6	Surroundings	16	8	0.667
f_7	Sound insulation	38	92	0.292
f_8	Service Characteristics	81	108	0.429
f_9	Service Experience	17	29	0.370
f_{10}	Room Types	15	23	0.395
f_{11}	Pick-up Service	5	12	0.294
f_{12}	Hotel Facilities	15	22	0.405

Table 8
The specific CRs.

CR_i	Topics	The specific CRs
CR_1	Acceptance of Opinions	Accept feedback and improve them
CR_2	Floor Characteristics	Room floors should be distinctive
CR_3	Way of Check-in	Convenient, fast, and diversified ways of check-in
CR_5	Hotel Environment	The hotel environment should be good
CR_6	Comprehensive Evaluation	A comprehensive evaluation should be high
CR_7	Surroundings	The scenery and environment of the hotel should be good
CR_8	Sound insulation	Soundproof effect quality should be good
CR_9	Service Characteristics	Service should be attentive, comfortable, and distinctive
CR_{10}	Service Experience	Service experience should be good, and staff attitude should be good
CR_{11}	Room Types	The room type and the facilities inside should be rich
CR_{12}	Pick-up Service	Provide pick-up service
CR_i	Hotel Facilities	Hotel facilities need to be diverse

According to the above analysis, Table 9 shows the final Kano category results for all topics.

By combining all the results from above, we could make an IPA-KANO category results figure to show the final priority, which is shown in Fig. 16. It presents the results of the integrated IPA-Kano category analysis for the eleven identified customer requirements. Each CR is simultaneously evaluated through the lenses of Kano classification, importance level, performance level, and its mapped IPA strategy, resulting in a synthesized strategic priority recommendation.

Step 9: Identify the IPA-Kano categories of CRs.

To better analyze the enhancement strategy of hotel customer demand, this paper integrates the IPA analysis model and the KANO model to construct the IPA-KNAO analysis matrix to analyze the attribute characteristics. The mean values of importance (0.083) and satisfaction (0.412). Since A_4 is classified as A, we do not discuss it here. The IPA-Kano category results are calculated as follows.

Note: HA: high attractive quality; LA: low attractive quality; LV: low value-added quality; HM: high must-be quality; LM: low must-be quality.

Step 10: Identify the weights of CRs.

According to the IPA-Kano categories and TF-IDF results of every CR, combined with the strategic priority, we obtain the final

Table 9
The Kano category results.

CR_i	Keywords	Sentiment orientation	Kano categories
CR_1	2	Positive	O
CR_2	1	Negative	M
CR_3	1	Negative	M
CR_5	4	Neutral	I
CR_6	3	Positive	A
CR_7	1	Negative	M
CR_8	1	Negative	M
CR_9	1	Negative	M
CR_{10}	1	Negative	M
CR_{11}	1	Negative	M
CR_{12}	1	Negative	M
CR_i	1	Negative	M

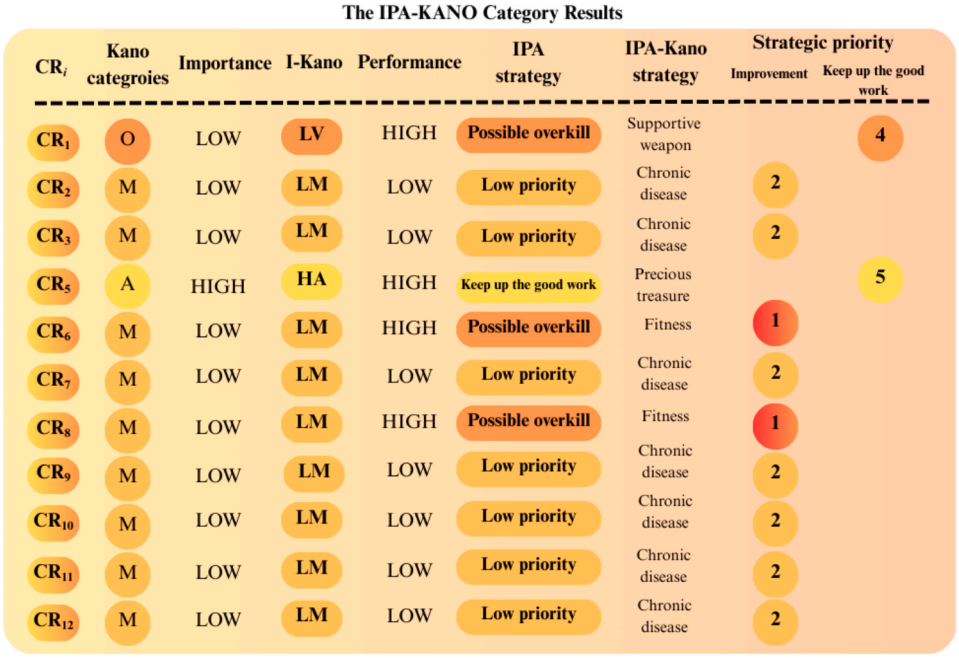


Fig. 16. The IPA-Kano category results.

weights $w_i (i = 1, 2, ..., 12)$ of CRs in Table 10.

Stage 2: Calculate the weight of PCs within the IVIF-QFD model.

Step 1: We invited a decision-making group that included three experts from universities and the government in hotel management. We obtained several PCs corresponding to hotel operation improvement according to the proposed CRs. The weights of every expert

Table 10
The Kano category results.

CR_i	Weights (w_i)
CR_1	0.029
CR_2	0.059
CR_3	0.132
CR_5	0.015
CR_6	0.147
CR_7	0.118
CR_8	0.162
CR_9	0.103
CR_{10}	0.088
CR_{11}	0.059
CR_{12}	0.088

are $\lambda_i(i = 1,2,3) = 0.333, 0.333, 0.333$. The specific PCs are shown as follows.

To enhance the proposed QFD framework's practical applicability, we mapped each PC to representative operational strategies based on expert interviews and industry best practices. This linkage, illustrated in Table 11, aims to guide hotel managers in effectively implementing the prioritized improvements.

Step 2: Construct the IVIF relationship matrix between Customer Requirements (CRs) and Product Characteristics (PCs) within the QFD framework. A decision-making expert group assesses the correlations between CRs and PCs and provides their evaluations using IVIF judgment. These evaluations are detailed in Figs. 17-19.

To evaluate the reliability of expert judgments in the QFD matrix, we conducted a Cohen's Kappa analysis based on discretized CR–PC scores from three independent domain experts. The analysis is shown in Appendix 4.

Step 3: The IVIF weights of PCs are obtained through Eq. (21). The IVIF weight of PC_1 based on Expert 1 can be calculated as follows.

$$\begin{aligned} TW_1 &= w_1 \times \widetilde{R}_{11} + w_2 \times \widetilde{R}_{21} + w_3 \times \widetilde{R}_{31} + w_5 \times \widetilde{R}_{51} + w_6 \times \widetilde{R}_{61} + w_7 \times \widetilde{R}_{71} + w_8 \times \widetilde{R}_{81} + w_9 \times \widetilde{R}_{91} + w_{10} \times \widetilde{R}_{101} \\ &\quad + w_{11} \times \widetilde{R}_{111} + w_{12} \times \widetilde{R}_{121} \\ &= 0.052 + 0.044 + 0.046 + 0.044 + 0.044 + 0.044 + 0.046 + 0.046 + 0.044 + 0.044 + 0.044 \\ &= 0.496 \end{aligned}$$

Next, the PCs' IVIF weights are defuzzified according to Eq. (17). The weights derived from experts A to C, presented in Tables 12-14, serve as inputs for the IVIF-VIKOR method. The weights shown in Table 15 will also be used as inputs for the IVIF-VIKOR approach.

Stage 3: Rank the alternatives using the IVIF-VIKOR model

Step 1. Build the assessment matrix for the alternatives.

The decision-making group evaluates the selected alternatives, and their specific interpretations are presented in Table 16. The corresponding assessment matrix is shown in Table 17.

Step 2. Convert the linguistic term into IVIF values.

In the assessment matrix, qualitative evaluations are transformed into IVIFNs using the linguistic scales provided in Fig. 10. The evaluations for each alternative are displayed in Table 18. The exact IVIF aggregation process is shown in Appendix 5.

Step 3. Rank the alternatives.

Calculate the best solution F^+ and the worst solution F^- based on PCs, and shown as follows. The conversion of IVIF into real scores is shown in Appendix 6.

② Group utility and individual regret values for IVIFNs are $S(A_k)$ in Eq. (28) and $R(A_k)$ in Eq. (29), for interval value are $\left[S_k^L, S_k^U\right]$ in Eq. (31) and (32), and $\left[R_k^L, R_k^U\right]$ in Eq. (33) and (34), respectively. The result is presented in Table 20 as follows:

③ Obtain the values of $Q_k = \left[Q_k^L, Q_k^U\right]$ according to Eq. (35) and (36).The parameter $\eta(0 < \eta \leq 1)$ is the optimal level of DM, with rational DM $\eta = 0.50$.The results are shown $S^* = 0.73, S^- = 3.79, R^* = 0.31, R^- = 1.39$.

Rank the alternative according to the ranking rules in Table 21, and the result is shown as follows:

$Q_5 < Q_2 < Q_3 > Q_1 < Q_4$; so,$A_5 > A_2 > A_3 > A_1 > A_4$.

In this regard, Knowledge & technology are selected as the most important alternatives for hoteliers to improve their hotel operations, and the remaining alternatives are ranked as Money, Human Resources, Time, and Materials & Equipment, respectively. Therefore, the hotel should pay attention to the investment in Knowledge & technology. Specifically, knowledge-related investments may include: (1) Staff training programs (e.g., service recovery, digital literacy); (2) Knowledge management systems (e.g., best-practice sharing platforms); (3) Guest preference databases (e.g., CRM-based personalization strategies). Technology-related investments may include: (1) Front-end technologies: mobile check-in/check-out systems, digital concierge apps, chatbot services; (2) Operational technologies: IoT-based room automation, smart energy systems, predictive maintenance tools; (3) Analytical technologies: sentiment analysis platforms, demand forecasting models. We summarize these examples, which were derived from current hotel management practices and expert consultations.

6. Comparative analysis

This section provides a comparative analysis to demonstrate the proposed method.

6.1. Comparison with traditional questionnaire survey methods

Traditional QFD methods rely on questionnaires and expert evaluations to determine CRs and their corresponding weights, a process that cannot be applied directly to our case. To highlight the unique aspects of our approach, we present comparisons with existing QFD methods.

Comparison with [39–41]: Agrawal [39] applied the balanced scorecard in conjunction with QFD to explore design targets for enhancing performance. Li et al. [40] used a new approach in which fuzzy QFD is combined with an extended probabilistic language model to calculate the importance of PCs in the context of CRs for the open design process. Liu [41] proposed a fuzzy QFD model for product design and selection. However, none of these studies incorporated in-depth data mining techniques to extract

Table 11
The Meaning of Specific PCs.

PCs	**Meanings**	**Example Actionable Strategies**
PC_1	Taking the customer's advice	- Implement a digital feedback dashboard - Introduce a monthly review-response system
PC_2	Designing comfortable room types	- Provide adjustable lighting and climate controls - Introduce ergonomic furniture layouts
PC_3	Enhancing the variety and convenience of check-in options	- Install self-check-in kiosks - Enable mobile check-in via app
PC_4	Enhancement of hotel surroundings	- Add green landscaping or rooftop gardens - Improve exterior signage and lighting
PC_5	Considering the location of the hotel	- Provide neighborhood maps and local guides - Partner with local attractions or transport
PC_6	Improvement of soundproofing of rooms	- Install double-glazed windows - Add acoustic panels or sound-insulating curtains
PC_7	Enhancement and improvement of hotel services	- Provide multilingual concierge services - Train staff in service recovery protocols
PC_8	Improvement of in-room amenities	- Upgrade Wi-Fi and smart TVs - Provide high-quality toiletries and bedding
PC_9	Provide personalized service	- Use CRM systems to remember guest preferences - Offer personalized welcome messages
PC_{10}	Improvement of hotel facilities	- Renovate gym/spa areas - Add coworking spaces or business lounges

A_i	PC_1	PC_2	PC_3	PC_4	PC_5	PC_6	PC_7	PC_8	PC_9	PC_{10}
A_1	AH	AE	MH	AL	VL	MH	VH	L	MH	VL
A_2	VL	AH	ML	AL	VL	VH	VL	ML	VL	VL
A_3	AE	L	AH	L	ML	L	VL	ML	L	L
A_5	AL	L	L	AH	VH	VL	VL	ML	AL	MH
A_6	AL	VL	VL	VH	AH	VL	AE	L	AL	MH
A_7	AL	VL	L	MH	ML	AH	ML	H	AL	H
A_8	AE	AL	ML	AL	AL	L	AH	ML	H	ML
A_9	AE	AL	L	AL	AL	L	AH	ML	H	ML
A_{10}	VL	H	L	AL	AL	VH	MH	AH	MH	H
A_{11}	ML	VL	AL	AE	L	AL	VH	MH	AH	H
A_{12}	VL	L	VL	AL	AL	MH	MH	VH	H	AH

Fig. 17. Relationship matrix between CRs and PCs based on Expert A.

A_i	PC_1	PC_2	PC_3	PC_4	PC_5	PC_6	PC_7	PC_8	PC_9	PC_{10}
A_1	AH	AE	MH	AL	VL	MH	VH	L	MH	VL
A_2	VL	AH	ML	AL	AL	VH	VL	ML	VL	AE
A_3	AE	L	AH	L	ML	AL	AL	AL	L	L
A_5	L	L	L	AH	VH	VL	AL	AL	AL	MH
A_6	AL	VL	VL	VH	AH	VL	AE	ML	AL	MH
A_7	L	L	L	MH	ML	AH	ML	H	AL	VH
A_8	AE	AL	ML	AL	AL	L	AH	ML	AH	ML
A_9	AE	AL	ML	AL	AL	AL	AH	ML	H	ML
A_{10}	VL	H	L	AL	AL	VH	MH	AH	MH	MH
A_{11}	ML	VL	AL	ML	AE	AL	VH	MH	AH	A
A_{12}	VL	L	VL	AL	AL	MH	MH	VH	H	AH

Fig. 18. Relationship matrix between CRs and PCs based on Expert B.

CRs directly. In contrast, our method employs a web crawler to extract CRs from online hotel reviews, enabling us to capture real-time customer insights and gain a competitive edge in the marketplace.

Zoghi et al. [42] integrated the Kano model with the analytic hierarchy process (AHP) and TOPSIS to facilitate material selection in deconstruction-oriented design within a fuzzy context. Shao et al. [43] introduced a three-stage MCDM approach that synergizes fuzzy logic, the Kano model, and failure mode and effects analysis (FMEA) for robust evaluations. He et al. [21] established a Kano-QFD-DEMATEL framework aimed at strengthening risk resilience in sustainable supply chain systems. Lizarelli et al. [44] applied a combination of the Kano model, service quality metrics, and QFD to advance service design practices. All of these papers determine CRs' Kano categories by questionnaire and obtain the weight of CRs. However, the questionnaire has been outmoded, so this paper uses the TF-IDF algorithm with the IPA model to eliminate the dependence on the traditional Kano questionnaire and classify CRs in a data-driven method. In addition, using a crawler and word frequency statistics, the Kano model can identify CRs in many online hotel comments.

A_i	PC_1	PC_2	PC_3	PC_4	PC_5	PC_6	PC_7	PC_8	PC_9	PC_{10}
A_1	AH	AE	MH	AL	VL	MH	VH	AL	MH	VL
A_2	VL	AH	ML	AL	VL	VH	AL	ML	VL	VL
A_3	AE	L	AH	ML	ML	ML	VL	AL	AL	AL
A_5	AL	L	AL	AH	AE	VL	AL	AL	AL	AE
A_6	AL	VL	VL	VH	AH	VL	AL	AL	AL	AE
A_7	AL	VL	L	MH	ML	AH	ML	H	AL	H
A_8	AE	AL	ML	AL	AL	AL	AH	ML	H	ML
A_9	AE	AL	AL	AL	AL	AL	AH	ML	H	ML
A_{10}	VL	H	AL	AL	AL	VH	MH	AH	AL	MH
A_{11}	ML	VL	AL	AE	ML	AL	VH	AE	AH	H
A_{12}	VL	AL	VL	ML	AL	MH	MH	VH	H	AH

Fig. 19. Relationship matrix between CRs and PCs based on Expert C.

Table 12
The weights of PCs based on Expert A.

PC_1	PC_2	PC_3	PC_4	PC_5	PC_6	PC_7	PC_8	PC_9	PC_{10}
0.099	0.099	0.098	0.100	0.099	0.101	0.103	0.100	0.100	0.100

Table 13
The weights of PCs based on Expert B.

PC_1	PC_2	PC_3	PC_4	PC_5	PC_6	PC_7	PC_8	PC_9	PC_{10}
0.099	0.098	0.098	0.099	0.099	0.101	0.103	0.100	0.101	0.102

Table 14
The weights of PCs based on Expert C.

PC_1	PC_2	PC_3	PC_4	PC_5	PC_6	PC_7	PC_8	PC_9	PC_{10}
0.099	0.099	0.098	0.100	0.098	0.101	0.103	0.100	0.100	0.101

Table 15
The weights of PCs for the IVIF-VIKOR method.

PC_1	PC_2	PC_3	PC_4	PC_5	PC_6	PC_7	PC_8	PC_9	PC_{10}
0.099	0.099	0.098	0.100	0.099	0.101	0.103	0.100	0.101	0.100

Table 16
Practical Investment Strategies for Hotel Operation Alternatives.

Alternative	Category	Example Practical Strategies
A1. Time	Resource Management	- Streamline check-in/out via automation - Minimize service wait times through scheduling AI
A2. Money	Financial Investment	- Allocate budget for high-ROI upgrades - Invest in loyalty program incentives
A3. Human Resources	Staff Development	- Provide frontline service training - Implement performance-based incentive systems
A4. Materials & Equipment	Physical Assets	- Upgrade in-room amenities (e.g., linens, furnishings) - Install energy-efficient equipment
A5. Knowledge & Technology	Strategic Capability	- Use IoT for room automation - Train staff on digital tools and service recovery - Implement CRM and data analytics platforms for personalization

6.2. Comparison with IVIF-QFD-TOPSIS methods

To validate the efficacy of the proposed approach, a comparative evaluation is conducted against an enhanced TOPSIS method combined with QFD, as proposed by Kilic et al. [45]. Recognized as a prevalent MCDM technique, TOPSIS determines optimal alternatives by measuring their closeness to the positive ideal solution and remoteness from the negative ideal counterpart [51]. In their

Table 17
The specific meanings of alternative A_i.

A_i	PC_1	PC_2	PC_3	PC_4	PC_5	PC_6	PC_7	PC_8	PC_9	PC_{10}
A_1	MH	MH	VL	AE	MH	ML	L	AE	L	VH
A_2	VL	ML	AE	VH	AH	AH	MH	VH	H	AH
A_3	VH	H	VH	AE	VL	VL	AH	AH	AH	VH
A_4	VL	AL	ML	AL	VL	H	VL	AH	L	AH
A_5	VH	AH	VH	H	H	VH	H	VH	H	VH

Table 18
Relationship matrix between A_i and PCs.

A_i	PC_1	PC_2	PC_3	PC_4	PC_5	PC_6	PC_7	PC_8	PC_9	PC_{10}
A_1	([0.5,0.7], [0.1,0.3])	([0.5,0.7], [0.1,0.3])	([0.1,0.3], [0.4,0.7])	([0.4,0.6], [0.2,0.4])	([0.5,0.7], [0.1,0.3])	([0.3,0.5], [0.2,0.5])	([0.2,0.4], [0.3,0.6])	([0.4,0.6], [0.2,0.4])	([0.2,0.4], [0.3,0.6])	([0.7,0.9], [0,0.1])
A_2	([0.1,0.3], [0.4,0.7])	([0.3,0.5], [0.2,0.5])	([0.4,0.6], [0.2,0.4])	([0.7,0.9], [0,0.1])	([0.8,1.0], [0,0])	([0.8,1.0], [0,0])	([0.5,0.7], [0.1,0.3])	([0.7,0.9], [0,0.1])	([0.6,0.8], [0,0.2])	([0.8,1.0], [0,0])
A_3	([0.7,0.9], [0,0.1])	([0.6,0.8], [0,0.2])	([0.7,0.9], [0,0.1])	([0.4,0.6], [0.2,0.4])	([0.1,0.3], [0.4,0.7])	([0.1,0.3], [0.4,0.7])	([0.8,1.0], [0,0])	([0.8,1.0], [0,0])	([0.8,1.0], [0,0])	([0.7,0.9], [0,0.1])
A_4	([0.1,0.3], [0.4,0.7])	([0,0.2], [0.5,0.8])	([0.3,0.5], [0.2,0.5])	([0,0.2], [0.5,0.8])	([0.1,0.3], [0.4,0.7])	([0.6,0.8], [0,0.2])	([0.1,0.3], [0.4,0.7])	([0.8,1.0], [0,0])	([0.2,0.4], [0.3,0.6])	([0.8,1.0], [0,0])
A_5	([0.7,0.9], [0,0.1])	([0.8,1.0], [0,0])	([0.7,0.9], [0,0.1])	([0.6,0.8], [0,0.2])	([0.6,0.8], [0,0.2])	([0.7,0.9], [0,0.1])	([0.6,0.8], [0,0.2])	([0.7,0.9], [0,0.1])	([0.6,0.8], [0,0.2])	([0.7,0.9], [0,0.1])

Table 19
The best solution F^+ and the worst solution F^-.

	PC_1	PC_2	PC_3	PC_4	PC_5	PC_6	PC_7	PC_8	PC_9	PC_{10}
F^+	([0.7,0.9], [0,0.1])	([0.8,1.0], [0,0])	([0.7,0.9], [0,0.1])	([0.7,0.9], [0,0.1])	([0.8,1.0], [0,0])	([0.8,1.0], [0,0])	([0.8,1.0], [0,0])	([0.8,1.0], [0,0])	([0.8,1.0], [0,0])	([0.8,1.0], [0,0])
F^-	([0.1,0.3], [0.4,0.7])	([0,0.2], [0.5,0.8])	([0.1,0.3], [0.4,0.7])	([0.1,0.3], [0.4,0.7])	([0.1,0.3], [0.4,0.7])	([0.1,0.3], [0.4,0.7])	([0.6,0.8], [0,0.2])	([0.4,0.6], [0.2,0.4])	([0.2,0.4], [0.3,0.6])	([0.7,0.9], [0,0.1])

Table 20
S and R interval numbers.

A_i	$[S_k^L, S_k^U]$	$[R_k^L, R_k^U]$
A_1	[3.27,3.30]	[0.90,0.93]
A_2	[1.14,1.56]	[0.46,0.57]
A_3	[1.26,1.58]	[0.40,0.41]
A_4	[3.37,3.79]	[1.08,1.39]
A_5	[0.73,1.38]	[0.31,0.72]

Table 21
Q interval number $\eta = 0.50$.

A_i	$Q_k = [Q_k^L, Q_k^U]$
A_1	[0.688,0.707]
A_2	[0.136,0.256]
A_3	[0.128,0.185]
A_4	[0.788,1.000]
A_5	[0,0.296]

study, Kilic et al. [45] utilized an IVIF-TOPSIS framework to assess and choose the most suitable information systems within the hospitality sector, accounting for multiple evaluation dimensions and inherent uncertainties in the decision-making environment. To emphasize the strengths of the IVIF-QFD-VIKOR model, we perform a detailed comparison with the IVIF-QFD-TOPSIS method. The procedural steps of the IVIF-TOPSIS technique are detailed below.

Step 1: According to Table 19, we have already obtained the best solution F^+ and the worst solution F^- based on PCs.

Step 2: Calculate the separation measures based on Eq. (38) and Eq. (39).

$$D_i^{k+} = \sqrt{\frac{1}{2}\sum_{J=1}^{m} w_j \left[\left\{ \mu_{\overline{A}}^L(\overline{x}_{ki}) - \mu_{\overline{A}}^L(\overline{x}_i^*) \right\}^2 + \left\{ \mu_{\overline{A}}^U(\overline{x}_{ki}) - \mu_{\overline{A}}^U(\overline{x}_i^*) \right\}^2 + \left\{ \nu_{\overline{A}}^L(\overline{x}_{ki}) - \nu_{\overline{A}}^L(\overline{x}_i^*) \right\}^2 + \left\{ \nu_{\overline{A}}^U(\overline{x}_{ki}) - \nu_{\overline{A}}^U(\overline{x}_i^*) \right\}^2 \right]} \tag{38}$$

$$D_i^{k-} = \sqrt{\frac{1}{2}\sum_{J=1}^{m} w_j\left[\left\{\mu_{\tilde{A}}^{L}(\bar{x}_{ki}) - \mu_{\tilde{A}}^{L}(\bar{x}_i^-)\right\}^2 + \left\{\mu_{\tilde{A}}^{U}(\bar{x}_{ki}) - \mu_{\tilde{A}}^{U}(\bar{x}_i^-)\right\}^2 + \left\{\nu_{\tilde{A}}^{L}(\bar{x}_{ki}) - \nu_{\tilde{A}}^{L}(\bar{x}_i^-)\right\}^2 + \left\{\nu_{\tilde{A}}^{U}(\bar{x}_{ki}) - \nu_{\tilde{A}}^{U}(\bar{x}_i^-)\right\}^2\right]} \quad (39)$$

Step 3: Calculate the closeness coefficient.

$$U_i = \frac{D_i^{k-}}{D_i^{k+} + D_i^{k-}}, where 0 \le U_i \le 1. \quad (40)$$

All of the above results will be shown in Table 22.

As shown in Table 22, the final ranking remains consistent with the order A5 > A2 > A3 > A1 > A4. Notably, the top-ranked alternative from the IVIF-TOPSIS method matches the proposed approach, further supporting the model's robustness. A detailed analysis reveals that the first alternative, Knowledge & Technology, stands out as the most favorable choice, according to this study's results (Table 23).

6.3. Effectiveness of the proposed method based on comparative analysis

To evaluate the effectiveness of the proposed IVIF-QFD-VIKOR framework enhanced with the CKM model, we compared its outputs against those of conventional QFD approaches that lack sentiment-driven classification or fuzzy weighting mechanisms. The results demonstrate clear advantages: (1) The CKM model extracted CRs with higher semantic clarity, combining sentiment analysis with Kano and IPA dimensions. This led to more targeted identification of critical customer expectations. (2) The use of IVIF logic significantly reduced the sensitivity to expert subjectivity, producing more consistent and interpretable PC weightings compared to crisp QFD matrices. (3) The VIKOR-based ranking method offered balanced, compromise-based strategy selection, unlike single-criterion ranking methods, which are often overly rigid. In comparative tests, the proposed method yielded improvement strategies that aligned more closely with industry benchmarks and expert consensus, confirming both theoretical and practical validity.

In addition, we adopted VIKOR in combination with IVIF-QFD to improve robustness and fairness in decision-making. Compared with the IVIF-QFD-TOPSIS model, which ranks alternatives based solely on their geometric closeness to the ideal solution, IVIF-QFD-VIKOR introduces the concept of compromise ranking, balancing group utility and individual regret. This distinction is particularly valuable in real-world decision contexts, such as hotel strategy planning, where trade-offs are inevitable among cost, customer satisfaction, and technical feasibility. Our results show that VIKOR selects alternatives that are not only technically optimal but also more balanced, avoiding domination by a single criterion. This makes it more suitable for multi-stakeholder environments and for identifying strategies that are both practical and well-aligned with customer needs.

These comparative results affirm that the proposed method improves the rationality, robustness, and real-world applicability of service decision-making derived from user-generated content.

7. Conclusion

This study presents a novel QFD model designed to enhance hotel operations by analyzing online reviews. It integrates the traditional IPA and Kano models with machine learning techniques to identify CRs using big data and text analysis. Furthermore, a QFD-MCDM framework is applied to rank PCs, where the VIKOR method is combined with IVIF for more robust decision-making.

Online hotel reservations have been a popular way to convince life, so online hotel reviews have become essential data for learning the customers' needs and requirements. The traditional questionnaire method cannot understand the CRs thoroughly, so we use a crawler, machine learning, and sentiment analysis to deal with the data from online hotel reviews. Applying big data to a practical hotel operation problem is excellent progress. The new data-driven IVIF-QFD-MCDM approach could be used in hotel operations to adapt to a competitive market. Based on our research, we have some conclusions.

The QFD method is feasible for studying hotel service quality. The essence of QFD is to take customer needs into account when enterprises make quality decisions. The method is introduced into the study of hotel service quality so that every link of the hotel service process can map the voice of customers to help enterprises find the critical service elements to meet customer needs and provide a decision-making basis for improving service quality.

This paper eliminates the traditional questionnaire method and analyzes CRs using big data. We use TF-IDF, crawler tools, and sentiment analysis techniques to improve the Kano model so that it can quantitatively divide CRs and completely understand customer needs.

Table 22
Results of D_i^{k+}, D_i^{k-} and U_i.

	D_i^{k+}	D_i^{k-}	U_i
i = 1	0.567	0.395	0.411
i = 2	0.368	0.569	0.607
i = 3	0.500	0.556	0.526
i = 4	0.705	0.376	0.348
i = 5	0.254	0.664	0.723

Table 23
Inter-Expert Agreement Measured by Cohen's Kappa.

Expert Pair	Cohen's Kappa
Expert A vs Expert B	0.900
Expert A vs Expert C	0.964
Expert B vs Expert C	0.865
Mean Agreement	0.910

Combining the QFD method with MCDM can help hotel operations. The QFD-VIKOR approach can improve the sustainable performance of hoteliers and make enterprises stand out in markets with fierce competition. Construct a HoQ model to describe the relationship between the CRs and PCs, and use the VIKOR method to determine the priority of PCs. This approach is a flexible and easy way to learn customer needs and enhance the operation of enterprises.

However, there are still some problems we need to study deeply. For instance, this paper uses online hotel review data compiled by Dr. Songbo Tan for data quality assurance, but its timeliness is not strong enough. In the future, we should consider crawling the data from a specific hotel to enhance the relevance and timeliness of the work. In addition, the decision information will change during different periods. Thus, we should pay attention to a dynamic MCDM approach to solve the problems. In addition, we need to apply and develop the proposed method into a product platform and modular product design.

Finally, we present prospects for future research. Since the current study presents a robust framework for translating customer reviews into prioritized hotel improvement strategies, it assumes static customer preferences. In reality, customer needs may shift over time due to events such as pandemics or seasonal trends. To tackle this limitation, upcoming studies are encouraged to integrate Dynamic Topic Modeling (DTM) for identifying temporal variations in topic significance and leverage recurrent neural networks (RNNs), particularly Long Short-Term Memory (LSTM) architectures, to model sentiment dynamics over time effectively. Such enhancements could further improve the adaptability and real-time relevance of the proposed decision-making framework.

Statements and declarations

Ethical approval: This article does not contain any studies with human participants or animals performed by any of the authors.
Consent for publication: The manuscript is approved by all authors for publication.
Compliance with Ethical Standards.
(1) Disclosure of potential conflicts of interest.
We declare that we do have no commercial or associative interests that represent a conflict of interests in connection with this manuscript. There are no professional or other personal interests that can inappropriately influence our submitted work.
(2) Research involving human participants and/or animals.
This paper does not include any researches with human participants or animals performed by any of the authors.

CRediT authorship contribution statement

Peide Liu: Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Xinming Shi:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Yingcheng Xu:** Writing – review & editing, Visualization, Validation, Software, Methodology, Data curation. **Ran Dang:** Writing – review & editing, Visualization, Supervision, Resources, Methodology. **Yifan Wu:** Writing – review & editing, Visualization, Validation, Supervision, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix 1

Category	Example Terms (Chinese)	English Translation / Context	Source
Sentiment-related words	舒适,干净,噪音,拖延,态度	Comfort, Cleanliness, Noise, Delay, Attitude	Extracted from raw corpus
Hotel service descriptors	前台,卫生间,早餐,接机,服务员	Front desk, Bathroom, Breakfast, Shuttle, Staff	Raw Corpus & Sogou Cell Dictionary
Location-based terms	商圈,交通便利,市中心,景区	Business district, Convenient transport, City center, Scenic spot	Sogou Cell Dictionary
Experience-enhancing terms	安静,隔音,灯光,温度,氛围	Quiet, Soundproofing, Lighting, Temperature, Ambience	Extracted from raw corpus
Negative sentiment markers	脏乱,嘈杂,拖延, 无视,不耐烦	Messy, Noisy, Delay, Ignored, Impatient	Sentiment lexicon / manual coding

Appendix 2. Consistency measurement for Sentiment-Based Kano classification

To assess the reliability and internal consistency of the sentiment-based Kano classification, we conducted a consistency measurement analysis. Specifically, we compared the sentiment polarity labels extracted using the RoBERTa model with the final Kano category assignments based on our rule-based approach. To facilitate binary comparison, Kano classifications were grouped into two categories: Must-be (M) versus all other categories (A, O, I). Sentiment labels were similarly grouped into Positive and Negative categories.

Cohen's Kappa coefficient was computed to quantify the level of agreement between sentiment orientation and Kano assignment. As shown in Fig. A1, the Cohen's Kappa score is 1.00, indicating perfect agreement. The confusion matrix demonstrates that all negative sentiment reviews aligned with Must-be classifications, while all positive sentiment reviews aligned with performance-based or attractive features. This result validates the internal consistency of the proposed classification logic.

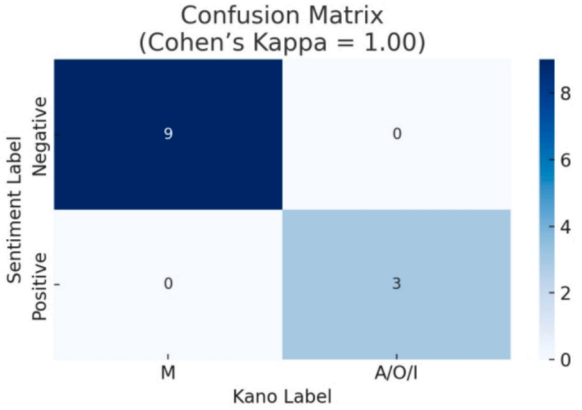


Fig. A1. Confusion Matrix for Sentiment Polarity vs. Kano Classification (Cohen's Kappa = 1.00) [25]

Appendix 3. Evaluation metrics of LDA models with varying topic numbers

Number of Topics	Perplexity	Variance	Coherence
2.0	−8.331113	0.17	0.472
3.0	−8.383335	0.39	0.414
4.0	−8.452440	0.34	0.420
5.0	−8.482878	0.34	0.419
6.0	−8.527893	0.33	0.397
7.0	−8.528300	0.32	0.542
8.0	−8.583490	0.30	0.419
9.0	−8.767436	0.29	0.513
10.0	−9.107279	0.29	0.469
11.0	−9.497116	0.27	0.513
12.0	−9.832490	0.24	0.581
13.0	−10.060820	0.23	0.535
14.0	−10.237250	0.22	0.553
15.0	−10.367710	0.22	0.563
16.0	−10.504560	0.19	0.499
17.0	−10.658720	0.20	0.521
18.0	−10.786890	0.20	0.536
19.0	−10.927500	0.18	0.504

(*continued on next page*)

(*continued*)

Number of Topics	Perplexity	Variance	Coherence
20.0	−11.059600	0.18	0.512
21.0	−11.216270	0.18	0.481
22.0	−11.362830	0.16	0.473
23.0	−11.490450	0.17	0.475
24.0	−11.617090	0.16	0.528
25.0	−11.751870	0.17	0.464
26.0	−11.890670	0.16	0.473
27.0	−12.027310	0.15	0.476
28.0	−12.166810	0.15	0.475
29.0	−12.304500	0.15	0.468
30.0	−12.468000	0.15	0.463

Appendix 4. Expert consistency analysis for CR–PC Scoring

To evaluate the consistency of expert judgments in the QFD matrix, we performed an inter-rater reliability analysis using Cohen's Kappa coefficient, a widely accepted statistical measure for assessing agreement between categorical ratings provided by multiple evaluators. The results indicate high consistency across all expert pairs, as summarized below:

These results demonstrate a very high level of agreement among the experts, exceeding the commonly accepted threshold for "almost perfect agreement" as per the Landis and Koch scale [52]. This provides strong empirical evidence supporting the reliability of our expert evaluations and mitigates concerns related to potential subjectivity. This validation step enhances the methodological transparency of our study and demonstrates that expert judgment was applied consistently and rigorously across evaluators.

Appendix 5. The real number scores of the relationship matrix between A_i and PCs

A_i	PC_1	PC_2	PC_3	PC_4	PC_5	PC_6	PC_7	PC_8	PC_9	PC_{10}
A_1	1.6	1.6	1.5	1.6	1.6	1.5	1.5	1.6	1.5	1.7
A_2	1.5	1.5	1.6	1.7	1.8	1.8	1.6	1.7	1.6	1.8
A_3	1.7	1.6	1.7	1.6	1.5	1.5	1.8	1.8	1.8	1.7
A_4	1.5	1.5	1.5	1.5	1.5	1.6	1.5	1.8	1.5	1.8
A_5	1.7	1.8	1.7	1.6	1.6	1.7	1.6	1.7	1.6	1.7

Appendix 6. The real number scores of the best solution F^+ and the worst solution F^-

	PC_1	PC_2	PC_3	PC_4	PC_5	PC_6	PC_7	PC_8	PC_9	PC_{10}
F^+	1.7	1.8	1.7	1.7	1.8	1.8	1.8	1.8	1.8	1.8
F^-	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.6	1.5	1.7

Data availability

No data was used for the research described in the article.

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